CMAI 480

Computer Vision

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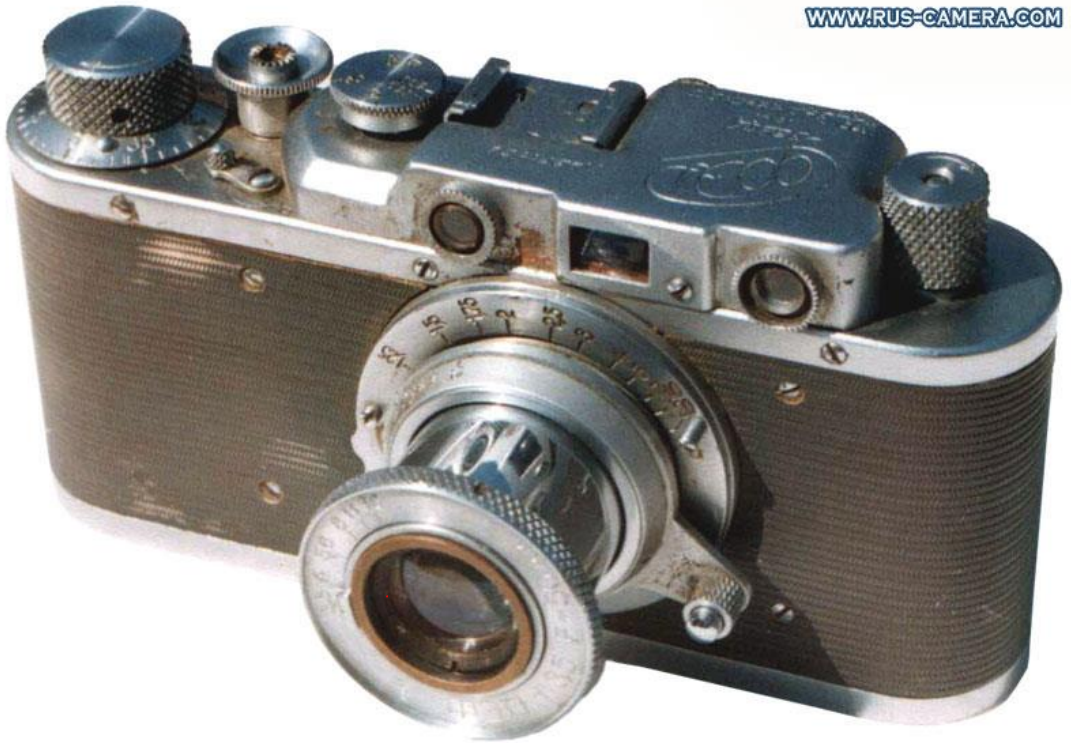
Outline

- Geometric primitives and transformations
  - Review of Linear Algebra
  - Projection geometry
    - Homogeneous Representation of 2D/3D entities or features
    - 2D/3D Image Transformations
- Human vision system
- Human vision for computer vision
- **Cameras and image formation**

Cameras

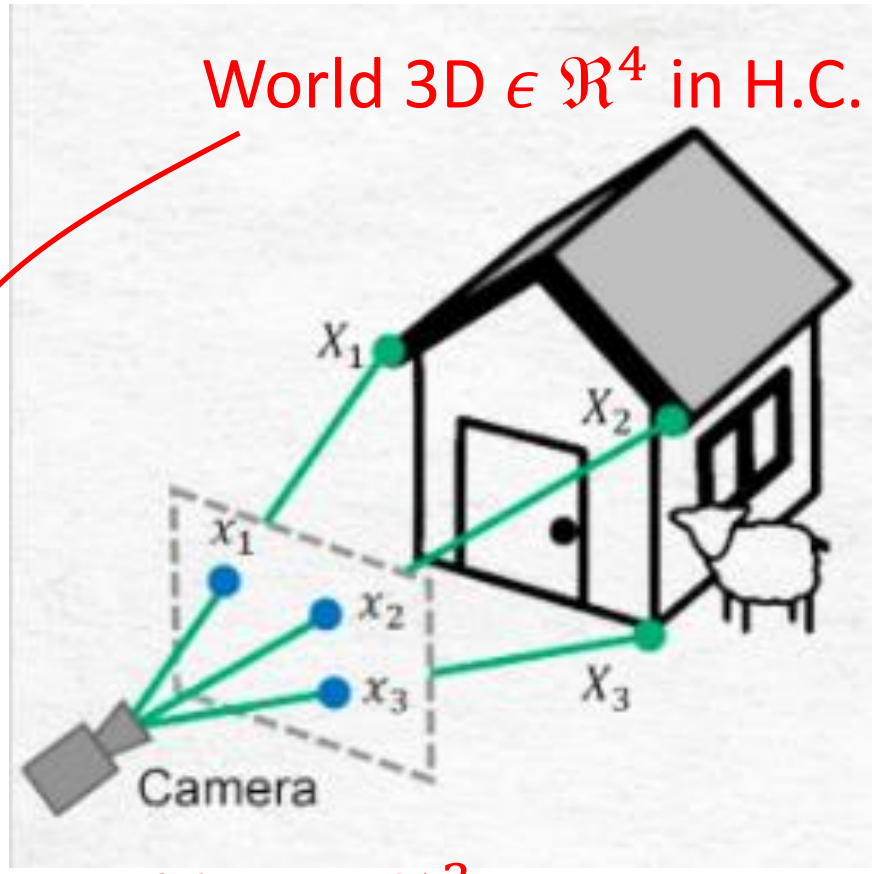
world around us

2D ← mapping / forward projection — 3D



$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}_{3\times1} = [\boldsymbol{Camera\ Matrix}]_{3\times4} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}_{4\times1}$$

Forward Projection or mapping

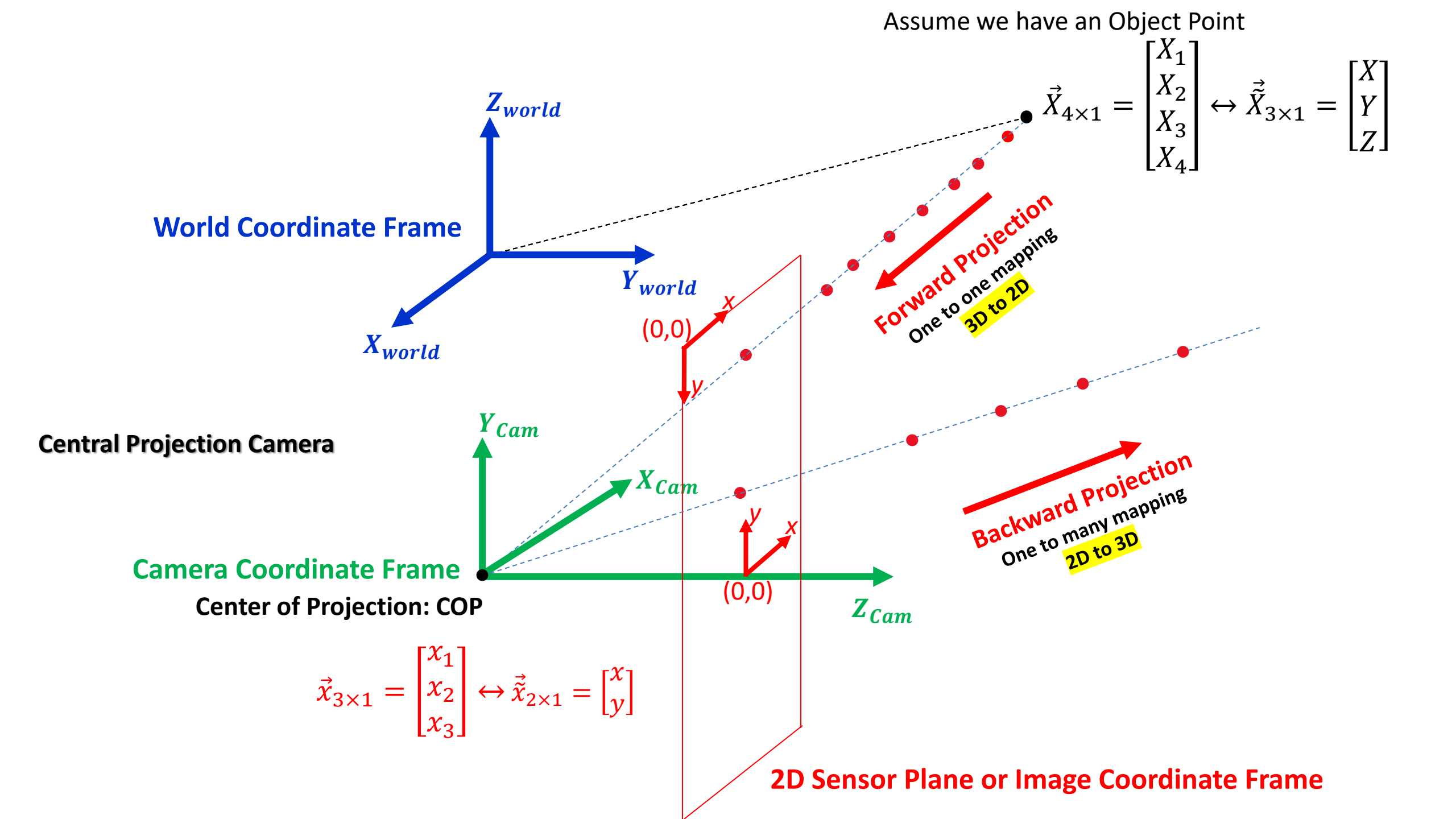


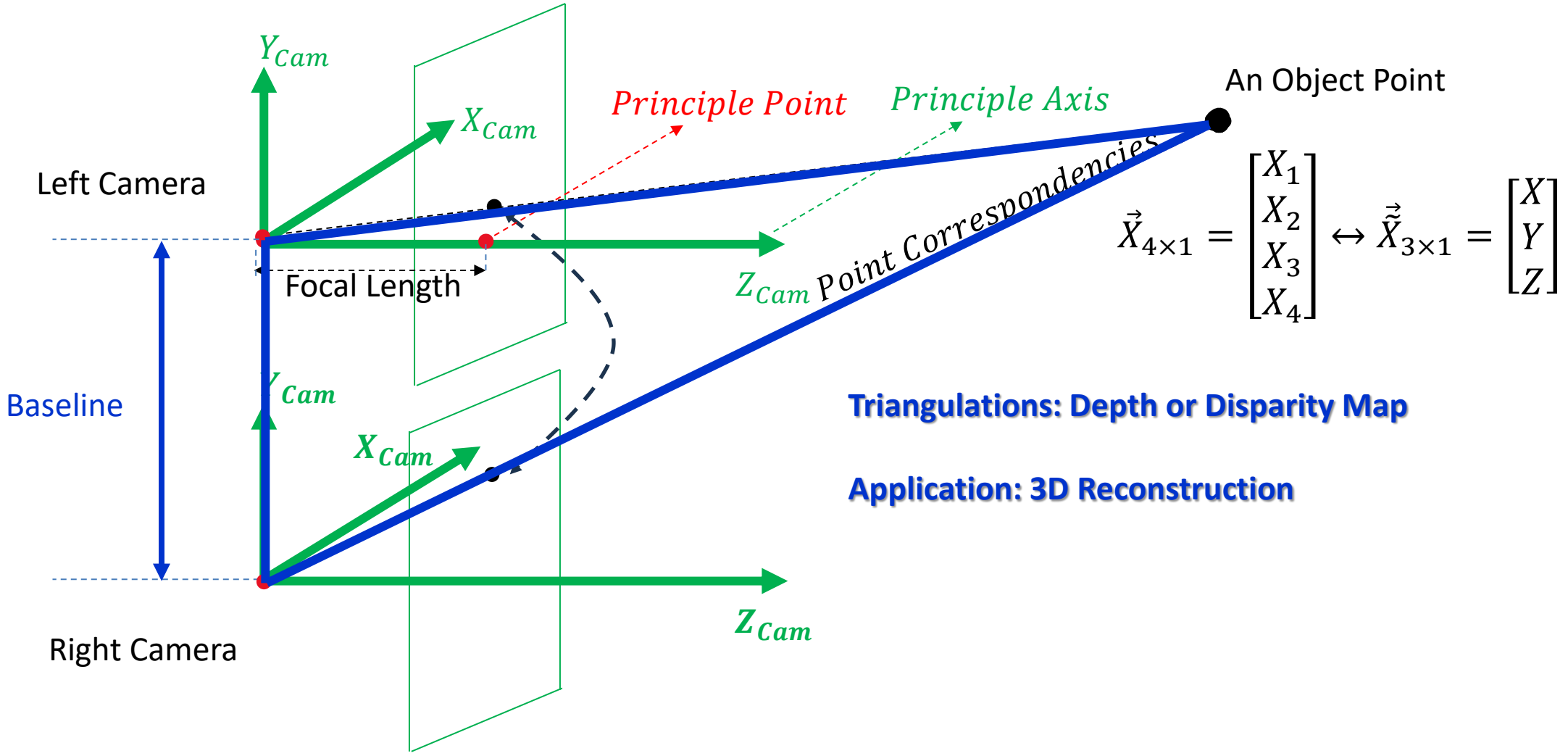
Our main job is now to derive such a very special Matrix $\boldsymbol{P_{3\times4}}$

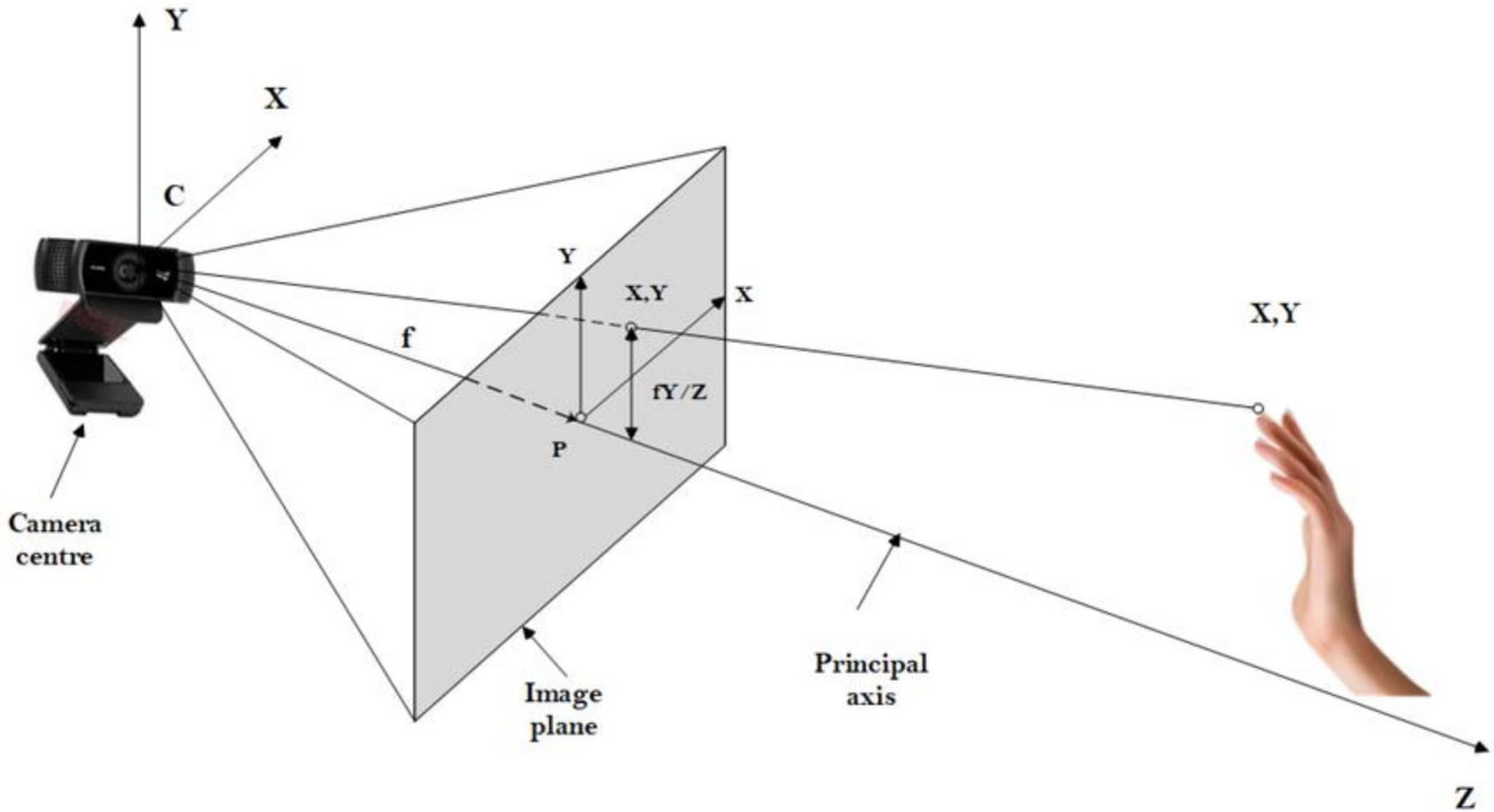
Focus: Central Projection Cameras

Fast Forward: Quick Observation $\boldsymbol{P_{3\times4}}$ is singular: Can't compute its inverse. However we can compute what is known as the **pseudoinverse** (most commonly the Moore–Penrose inverse), which is a generalization of the standard matrix inverse that exists for any rectangular or singular matrix **(Not unique)**.

$P^{\dagger}$:pseudo-inverse $P^{\dagger} = P^T(PP^T)^{-1}$
$PP^{\dagger} = I$

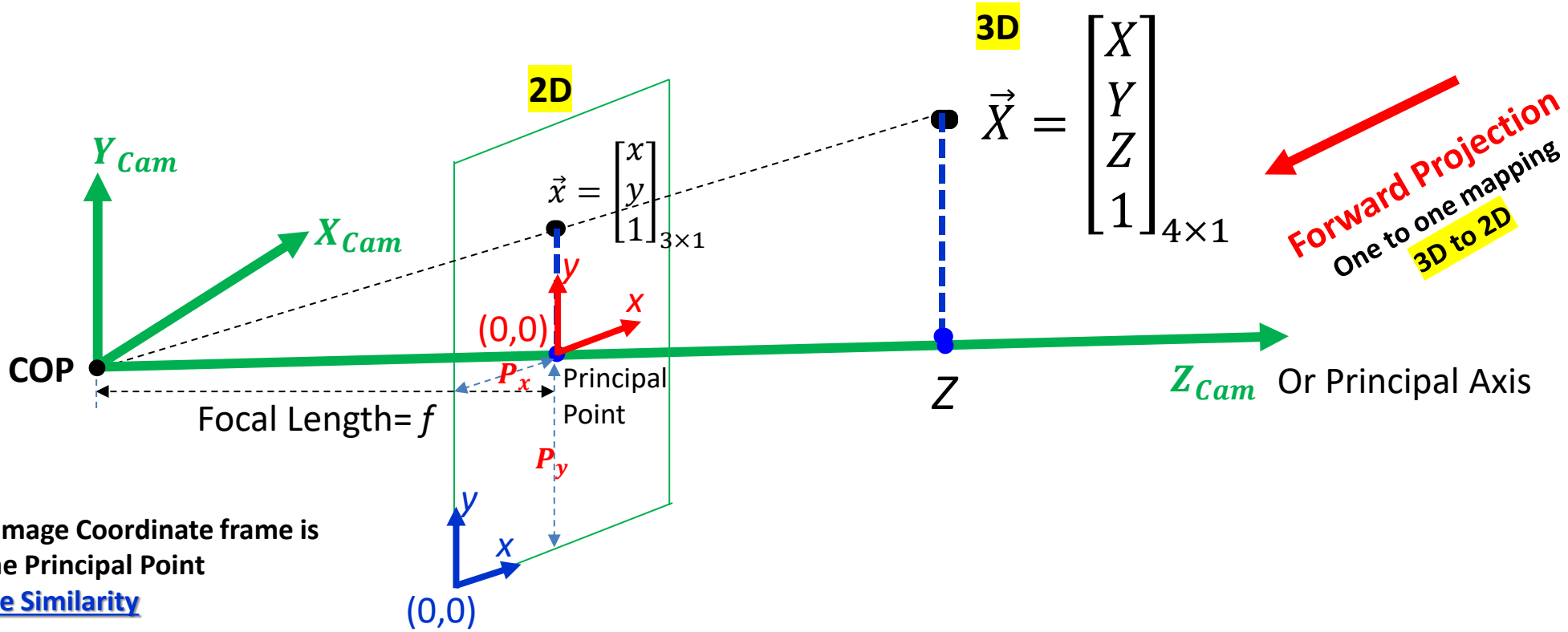

Assume we have an Object Point
$\vec{X}_{4\times1} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} \leftrightarrow \vec{\tilde{X}}_{3\times1} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$
Z_{world}
World Coordinate Frame
Y_{world}
X_{world}
Forward Projection
One to one mapping
3D to 2D
(0,0)
x
y
Central Projection Camera
Y_{Cam}
X_{Cam}
Backward Projection
One to many mapping
2D to 3D
Camera Coordinate Frame
Center of Projection: COP
(0,0)
Z_{Cam}
$\vec{x}_{3\times1} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leftrightarrow \vec{\tilde{x}}_{2\times1} = \begin{bmatrix} x \\ y \end{bmatrix}$
2D Sensor Plane or Image Coordinate Frame


Y_{Cam}
X_{Cam}
Principle Point
Principle Axis
An Object Point
Left Camera
Focal Length
Z_{Cam}
Point Correspondencies
$\vec{X}_{4\times1} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} \leftrightarrow \vec{\tilde{X}}_{3\times1} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$
Baseline
Y_{Cam}
X_{Cam}
Z_{Cam}
Right Camera
Triangulations: Depth or Disparity Map
Application: 3D Reconstruction


Y
X
C
Y
X,Y
X
f
fY/Z
P
X,Y
Camera
centre
Image
plane
Principal
axis
Z

Assumption:

- **Let the World Frame Coincides with the Camera Coordinate Frame**



Assume: The Image Coordinate frame is centered at the Principal Point

Apply: Triangle Similarity

$$\frac{y}{Y} = \frac{f}{Z}, \qquad \frac{x}{X} = \frac{f}{Z}$$

Or

$$y = \frac{fY}{Z}, \qquad x = \frac{fX}{Z}$$

Usually: 2D Sensor Plane is referenced to one of the corners of the Image Plane

Now assume: Image Coordinate frame is centered at one of the corners on the sensor plane

$$y = \frac{fY}{Z} + P_x, \qquad x = \frac{fX}{Z} + P_y$$

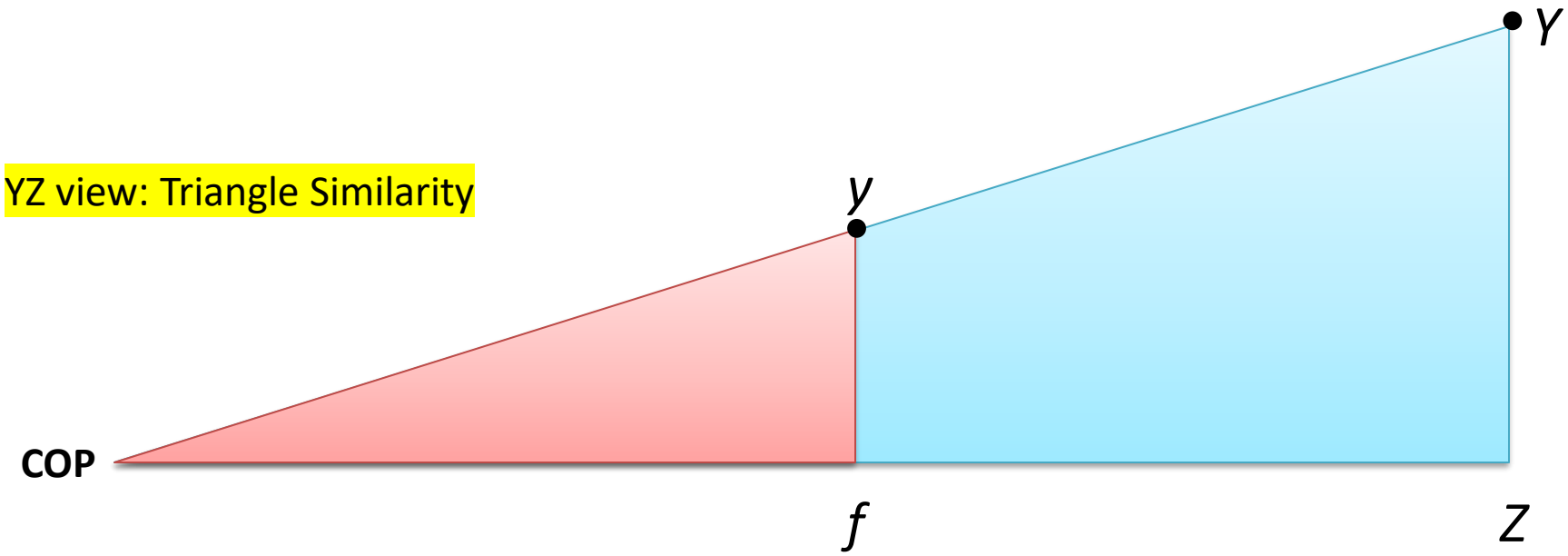
Or

$$y = \frac{fY + ZP_x}{Z}, \quad x = \frac{fX + ZP_y}{Z}$$

Or

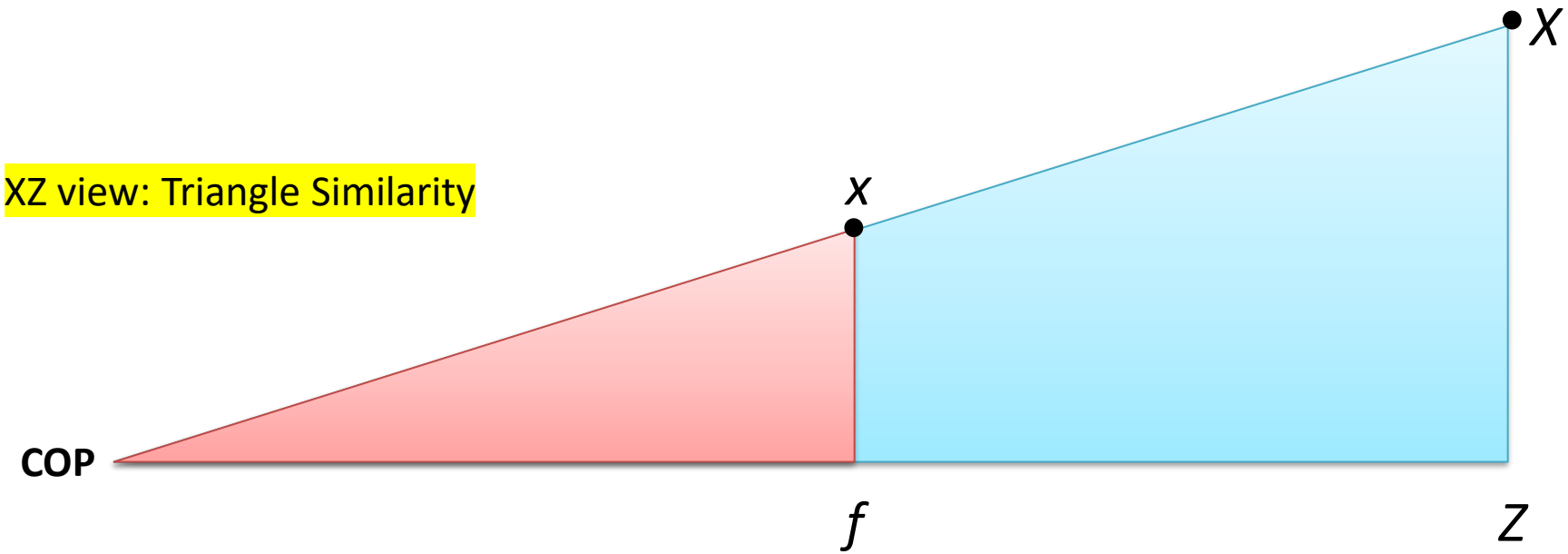
$$Zy = fY + ZP_x, \quad Zx = fX + ZP_y$$

YZ view: Triangle Similarity



$$\frac{y}{Y} = \frac{f}{Z}$$

XZ view: Triangle Similarity



$$\frac{x}{X} = \frac{f}{Z}$$

- Note: Even if the World Coordinate Frame is not to coincide with Camera World Frame, we can always enforce different settings through the use of the 3D Euclidian transformation or homography (**6 DoF**) which is given as follows:

$$\begin{bmatrix} X_{Cam} \\ Y_{Cam} \\ Z_{Cam} \\ 1 \end{bmatrix}_{4\times1} = \begin{bmatrix} \boldsymbol{r_{11}} & \boldsymbol{r_{12}} & \boldsymbol{r_{13}} & \boldsymbol{t_x} \\ \boldsymbol{r_{21}} & \boldsymbol{r_{22}} & \boldsymbol{r_{23}} & \boldsymbol{t_y} \\ \boldsymbol{r_{31}} & \boldsymbol{r_{32}} & \boldsymbol{r_{33}} & \boldsymbol{t_z} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{1} \end{bmatrix}_{4\times4} \begin{bmatrix} X_{world} \\ Y_{world} \\ Z_{world} \\ 1 \end{bmatrix}_{4\times1}$$

$$H_{4\times4} = \begin{bmatrix} R_{3\times3} & \vec{t}_{3\times1} \\ \vec{0}^T_{1\times3} & 1 \end{bmatrix}$$

$$Currently,\ H_{4\times4} = \begin{bmatrix} \boldsymbol{1} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{1} & \boldsymbol{0} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{1} & \boldsymbol{0} \\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{1} \end{bmatrix}$$

2D Image point $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$

3D object point $= \begin{bmatrix} X \\ Y \\ Z_1 \\ 1 \end{bmatrix}_{4\times1}$

$$x = \frac{fX + p_x Z_1}{Z_1}$$

$$y = \frac{fY + p_y Z_1}{Z_1}$$

physical

2D $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}_{3\times1} = \begin{bmatrix} xz \\ yz \\ z \end{bmatrix} = \begin{bmatrix} xZ_1 \\ yZ_1 \\ Z_1 \end{bmatrix} = \begin{bmatrix} fX + p_x Z_1 \\ fY + p_y Z_1 \\ Z_1 \end{bmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}_{3\times4} \cdot \begin{bmatrix} X \\ Y \\ Z_1 \\ 1 \end{bmatrix}_{4\times1}$ 3D

Camera matrix.

Homogeneous

3D ← 4D

$$\vec{x}_{3\times1} = P_{3\times4}\vec{X}_{Cam,World}$$

$$\vec{x}_{3\times1} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \vec{X}_{4\times1}$$

$P_{3\times4}$ = Camera Matrix

Augmented matrix

$$= K_{3\times3}\left[\, I_{3\times3} \mid \vec{0}_{3\times1} \,\right]_{3\times4}$$

$$= \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}_{3\times3} \left[\, R_{3\times3} \mid \vec{T}_{3\times1} \,\right]_{3\times4}$$

Non-singular & upper Triangle

called

Intrinsic parameters of the Camera

Extrinsic parameters of the Camera

of the camera with respect to the World Frame

Rotation

Translation

Note: if world frame coincides with camera frame **and image plane is centered at principal point**

$$\frac{x}{X} = \frac{f}{Z} \rightarrow x = \frac{fX}{Z} \Rightarrow xZ = fX$$

$$\frac{y}{Y} = \frac{f}{Z} \rightarrow y = \frac{fY}{Z} \Rightarrow yZ = fY$$

Assume $z = Z$

Triangle similarity

2D Image point in homogeneous

$$\begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} xz \\ yz \\ z \end{bmatrix} = \begin{bmatrix} xz \\ yz \\ z \end{bmatrix} = \begin{bmatrix} fX \\ fY \\ Z \end{bmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

3×1 (homogeneous) $\quad 3 \times 4 \quad 4 \times 1$

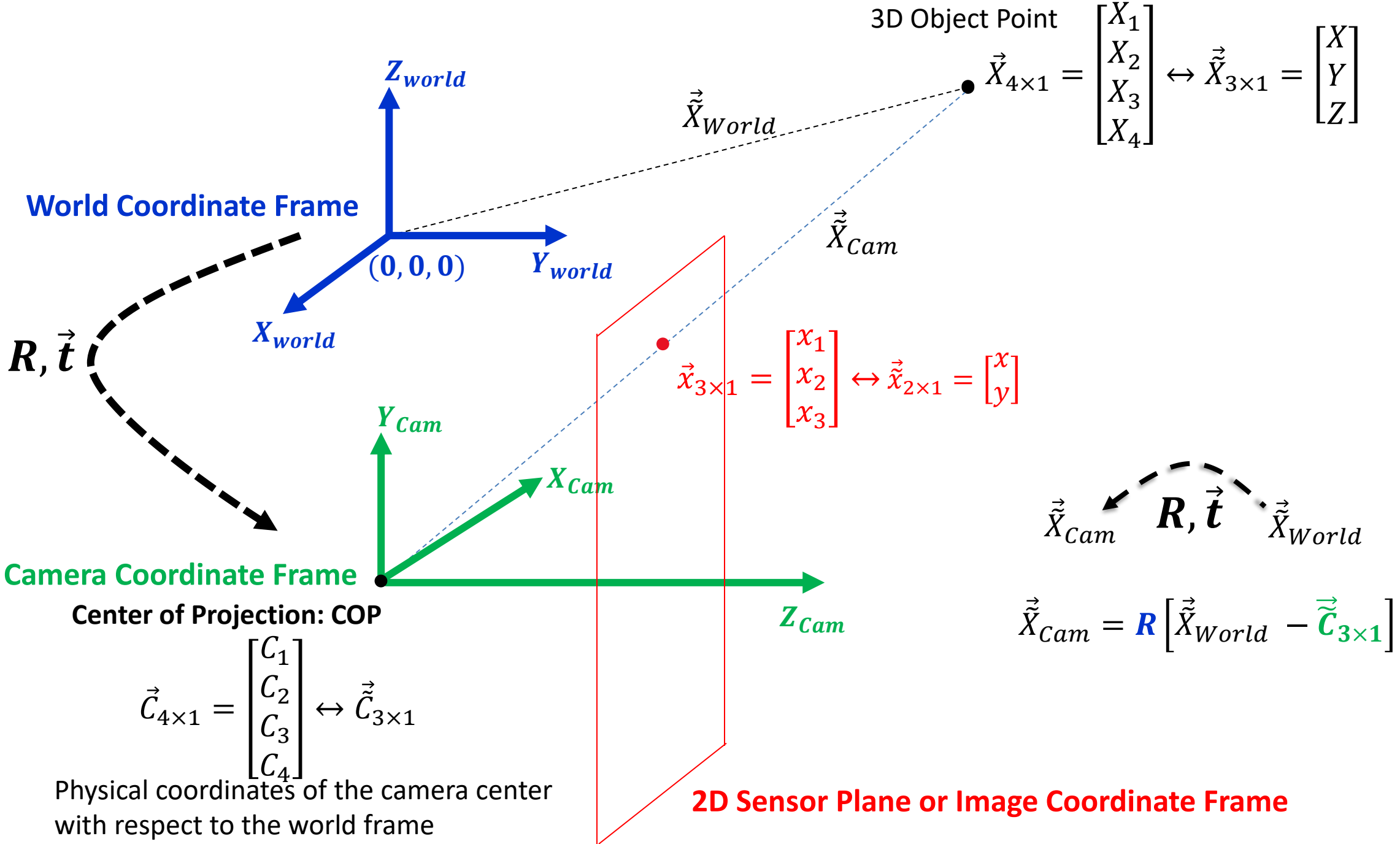
3D object in homogeneous

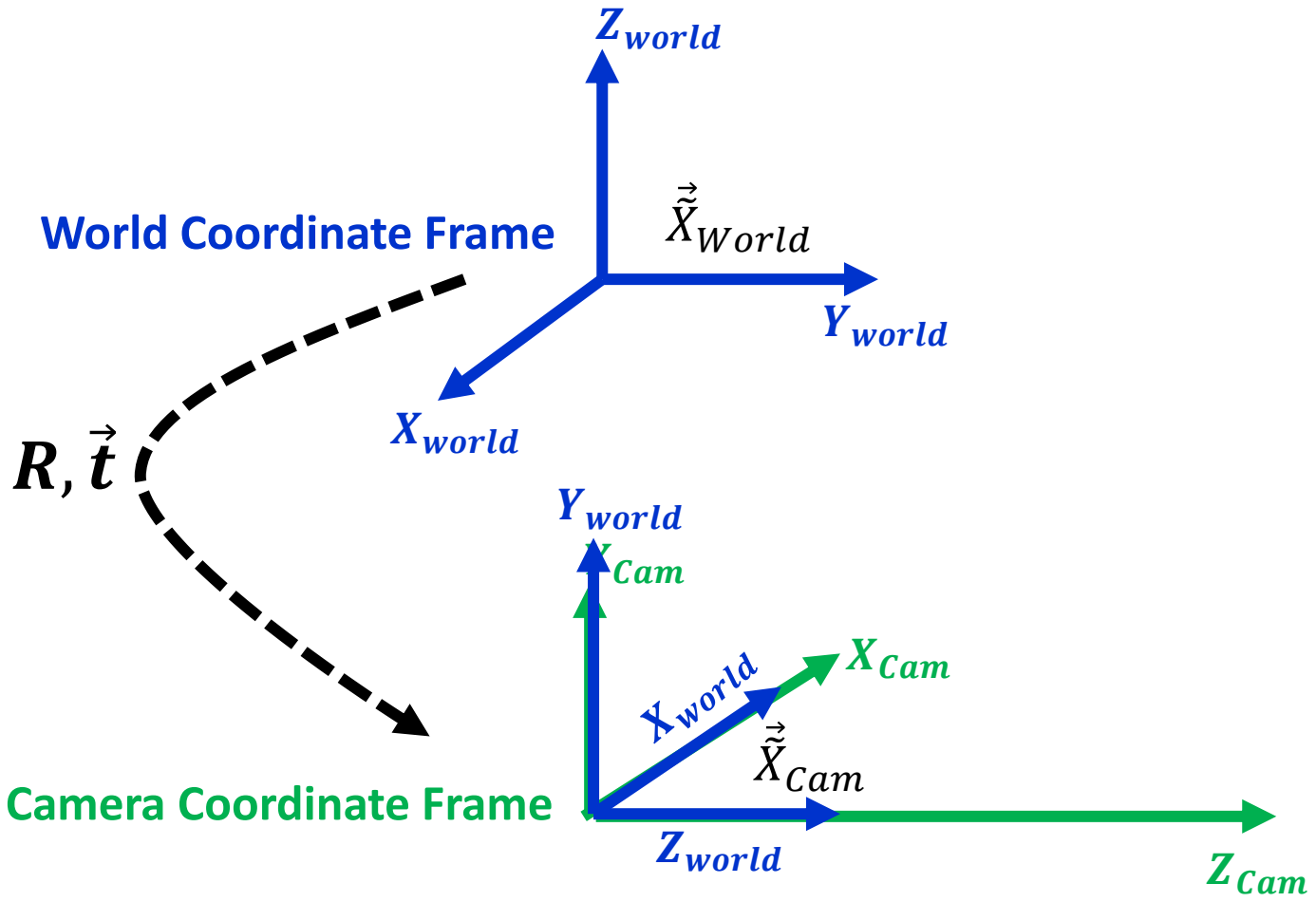
$$= \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3} \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}_{3\times4} = K_{3\times3} \underbrace{\left[I_{3\times3} \mid \vec{0}_{3\times1} \right]}_{\text{Augmented Matrix } 3\times4}$$

But, we will **assume 2D Sensor Plane is referenced to one of the corners of the Image Plane**

$$So, K_{3\times3} = \begin{bmatrix} f & 0 & P_x \\ 0 & f & P_y \\ 0 & 0 & 1 \end{bmatrix}, \text{not} \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- Let’s Now assume World Frame is different from Camera Frame


3D Object Point
$\vec{X}_{4\times1} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix} \leftrightarrow \vec{\tilde{X}}_{3\times1} = \begin{bmatrix} X \\ Y \\ Z \end{bmatrix}$
Z_{world}
$\vec{\tilde{X}}_{World}$
$\vec{\tilde{X}}_{Cam}$
World Coordinate Frame
$(0,0,0)$
Y_{world}
X_{world}
$R, \vec{t}$
$\vec{x}_{3\times1} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \leftrightarrow \vec{\tilde{x}}_{2\times1} = \begin{bmatrix} x \\ y \end{bmatrix}$
Y_{Cam}
X_{Cam}
$\vec{\tilde{X}}_{Cam}$ $R, \vec{t}$ $\vec{\tilde{X}}_{World}$
Camera Coordinate Frame
Z_{Cam}
Center of Projection: COP
$\vec{\tilde{X}}_{Cam} = R\left[\vec{\tilde{X}}_{World} - \vec{\tilde{C}}_{3\times1}\right]$
$\vec{C}_{4\times1} = \begin{bmatrix} C_1 \\ C_2 \\ C_3 \\ C_4 \end{bmatrix} \leftrightarrow \vec{\tilde{C}}_{3\times1}$
Physical coordinates of the camera center with respect to the world frame
2D Sensor Plane or Image Coordinate Frame



$$\vec{\vec{X}}_{Cam} = \left[\vec{\vec{X}}_{World} - \vec{\vec{C}}_{3\times1}\right]$$

$$\vec{\vec{X}}_{Cam} = \boldsymbol{R}\left[\vec{\vec{X}}_{World} - \vec{\vec{C}}_{3\times1}\right]$$

Alignment of the World Frame with the Camera Frame

$$\vec{\tilde{X}}_{Cam} = \boldsymbol{R}\left[\vec{\tilde{X}}_{World} - \vec{\tilde{C}}_{3\times1}\right]$$



$$\vec{\tilde{X}}_{Cam} = \boldsymbol{R}\,\vec{\tilde{X}}_{World} - \boldsymbol{R}\vec{\tilde{C}}_{3\times1}$$



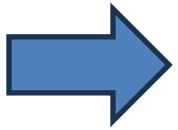
H.C.

$$\begin{bmatrix}\vec{\tilde{X}}_{Cam}\\ 1\end{bmatrix} = \begin{bmatrix}\boldsymbol{R} & -\boldsymbol{R}\vec{\tilde{C}}_{3\times1}\\ \vec{0}^{T}_{1\times3} & 1\end{bmatrix}_{4\times4}\begin{bmatrix}\vec{\tilde{X}}_{World}\\ 1\end{bmatrix}$$

Let $\vec{t}_{3\times1} = -\boldsymbol{R}\vec{\tilde{C}}_{3\times1}$



$$\begin{bmatrix}\vec{\tilde{X}}_{Cam}\\ 1\end{bmatrix} = \begin{bmatrix}\boldsymbol{R} & \vec{t}\\ \vec{0}^{T}_{1\times3} & 1\end{bmatrix}_{4\times4}\begin{bmatrix}\vec{\tilde{X}}_{World}\\ 1\end{bmatrix}$$



$$\vec{X}_{Cam} = \begin{bmatrix}\boldsymbol{R} & \vec{t}\\ \vec{0}^{T} & 1\end{bmatrix}_{4\times4}\vec{X}_{World}$$

$$\vec{X}_{Cam} = \begin{bmatrix} \boldsymbol{R} & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix}_{4\times4} \vec{X}_{World}$$

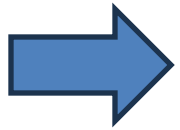
But we know that: $\vec{x}_{3\times1} = P_{3\times4}\vec{X}_{Cam}$



$$\vec{x}_{3\times1} = P_{3\times4} \begin{bmatrix} \boldsymbol{R} & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix}_{4\times4} \vec{X}_{World}$$



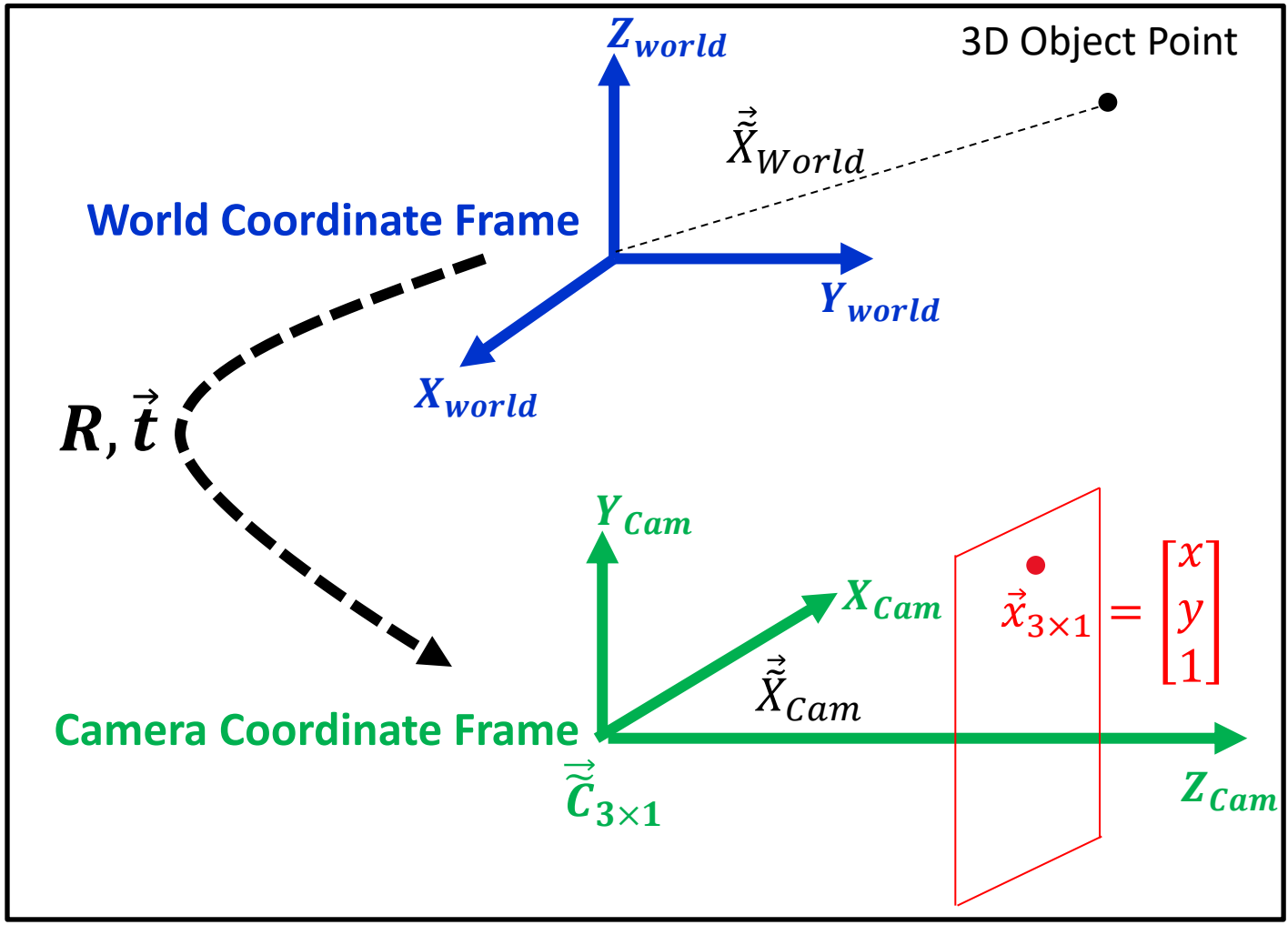
$$\vec{x}_{3\times1} = K_{3\times3}[I_{3\times3}|\vec{0}_{3\times1}]_{3\times4} \begin{bmatrix} \boldsymbol{R} & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix}_{4\times4} \vec{X}_{World}$$



New $P_{3\times4}$

$$\vec{x}_{3\times1} = K_{3\times3}[R_{3\times3}|\vec{t}_{3\times1}]_{3\times4}\vec{X}_{World}$$

Note: $\vec{t}_{3\times1} = -\boldsymbol{R}\vec{\tilde{C}}_{3\times1}$



Let's now go back to $K_{3\times3}$: Intrinsic Parameters:

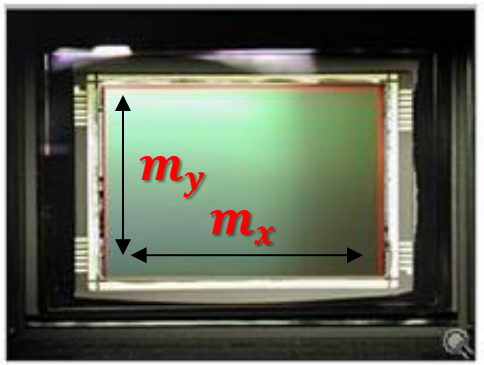
- Non-Singular and
- upper Triangle

$$K_{3\times3} = \begin{bmatrix} f & 0 & P_x \\ 0 & f & P_y \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow K_{3\times3} = \begin{bmatrix} \propto_x & 0 & x_0 \\ 0 & \propto_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

Considering the sampling of the image plane

Considering the sampling of the image plane



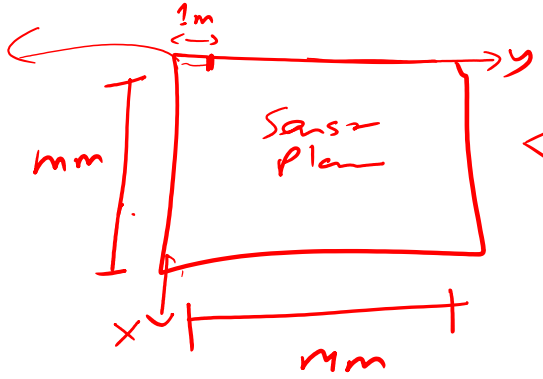




- So recall that we assumed the CCD image sensor have m_x elements per unit length along x and m_y elements per unit length along y.
- We may now associate the "**discretized**" location (I, J) with the pixel at (x, y), with:
  - $I = \boldsymbol{m_x} x$ and $J = \boldsymbol{m_y} y$
    - Note 1: I and J are known as the pixel coordinates in image **(True pixels)**
    - Note 2: Obviously, the products $m_x x$ and $m_y y$ do not by themselves yield integer values . However, to get integers indices associated with the cells, all we have to do is to ignore the fractional parts of $m_x x$ and $m_y y$.

$\begin{bmatrix} I \\ J \\ 1 \end{bmatrix}$ ⟷ establish a relationship $\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$ ✓

we need to deal with this True pixel coordinate



Rectangle $\alpha_x \neq \alpha_y$.

square $\alpha_x = \alpha_y$

every unit length in mm contains several samples along both x & y.

Let $\begin{bmatrix} I \\ J \\ 1 \end{bmatrix}$ be the true pixel coordinates

where

$I = m_x x$

$J = m_y y$

$x = \frac{fX}{Z_1} + p_x$

$y = \frac{fY}{Z_1} + p_y$

$$I = m_x\left[\frac{fX}{Z} + p_x\right], \qquad J = m_y\left(\frac{fY}{Z} + p_y\right)$$

$$K = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}$$

3 DoF

$$\begin{bmatrix} I \\ J \\ 1 \end{bmatrix} = \begin{bmatrix} m_x f & 0 & m_x p_x \\ 0 & m_y f & m_y p_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ 1 \end{bmatrix},$$

$$= \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$

Let $\alpha_x = m_x f$

$\alpha_y = m_y f$

$x_0 = m_x p_x$

$y_0 = m_y p_y$

our new K

K has 4 DoF

Now on, we will assume the true pixel coordinates $\begin{bmatrix} I \\ J \\ 1 \end{bmatrix}$ are represented by $\vec{x}_{3\times 1}$ in H.C.

Summary

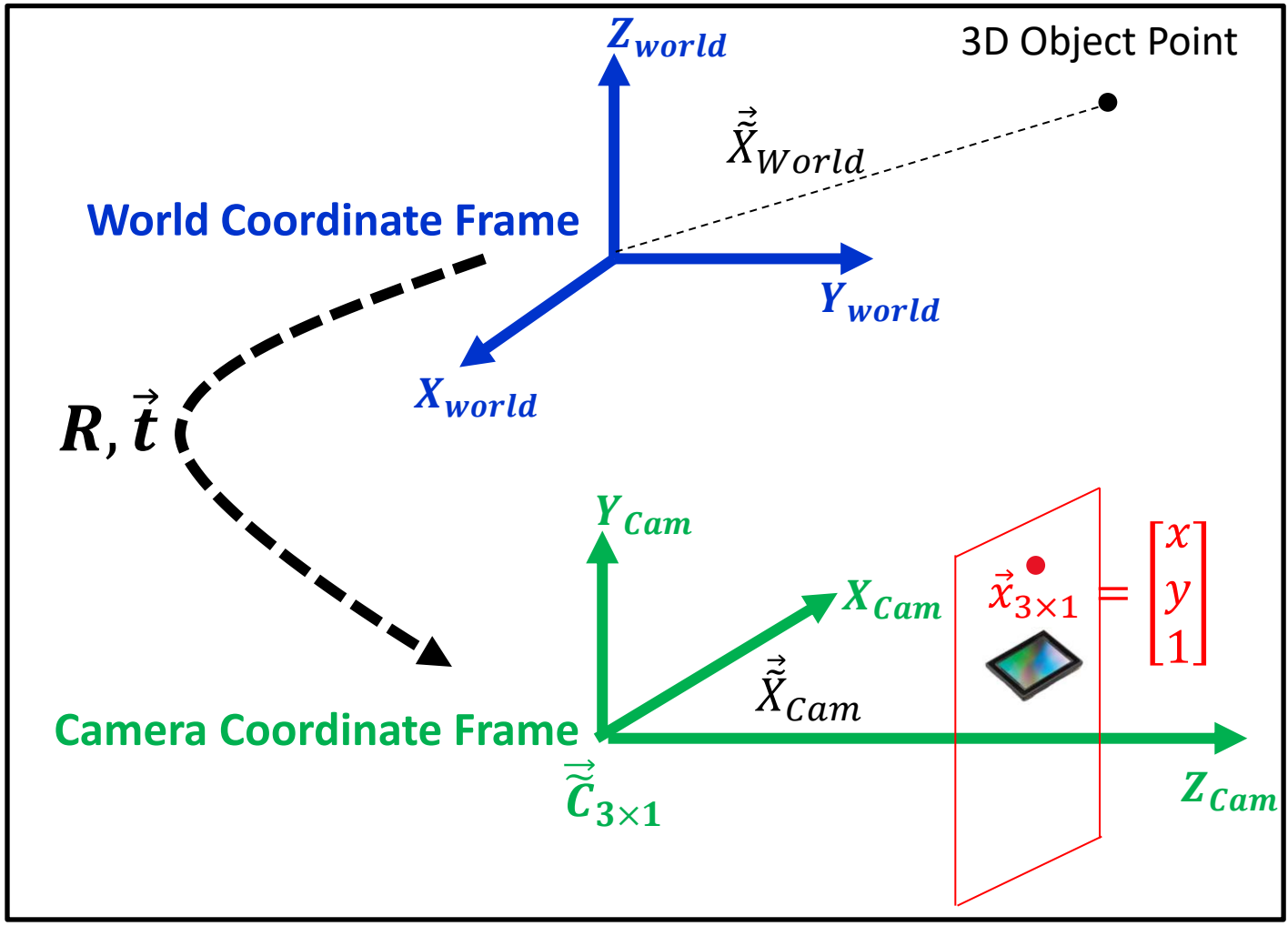
New $P_{3\times4}$

$$\vec{x}_{3\times1} = K_{3\times3}[R_{3\times3}|\vec{t}_{3\times1}]_{3\times4}\vec{X}_{World}$$

True pixel

$$\vec{t}_{3\times1} = -\boldsymbol{R}\vec{\tilde{C}}_{3\times1}$$

$$K_{3\times3} = \begin{bmatrix} \propto_x & 0 & x_0 \\ 0 & \propto_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$



Note:

- $\boldsymbol{P}$ has **11 DoF** (only ratios matter)
- $\boldsymbol{R}$ has 3 DoF
- $\vec{t}_{3\times1}$ has 3 DoF
- $K_{3\times3}$ has 4 DoF

We are missing one DoF, which is skew factor **s**

Ideally, **s = 0** (pinhole camera)

$$K_{3\times3} = \begin{bmatrix} \propto_x & \boldsymbol{s} & x_0 \\ 0 & \propto_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$$, So *K* now has 5 DOF

Outline

- Cameras Geometry and image formation

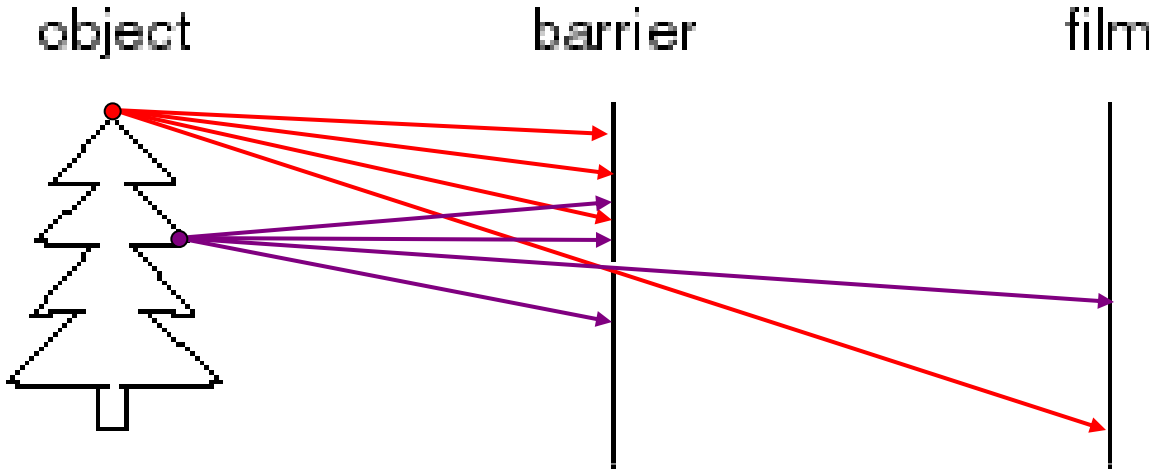
Cameras

- Recall:
  - A camera is a mapping from the 3D world to a 2D image.
  - Our primary focus will be on cameras that carry out Central Projection.
    - Central projection means that all rays that connect a world point with its corresponding image point pass through a common point known as the Center of Projection (CoP).

Camera Models

- At this time we are interested in two special cases of the central projection cameras:
  1. When the COP is at a finite location; and
  2. When the COP is at infinity
- A good place to start for understanding both is the basic pinhole camera.

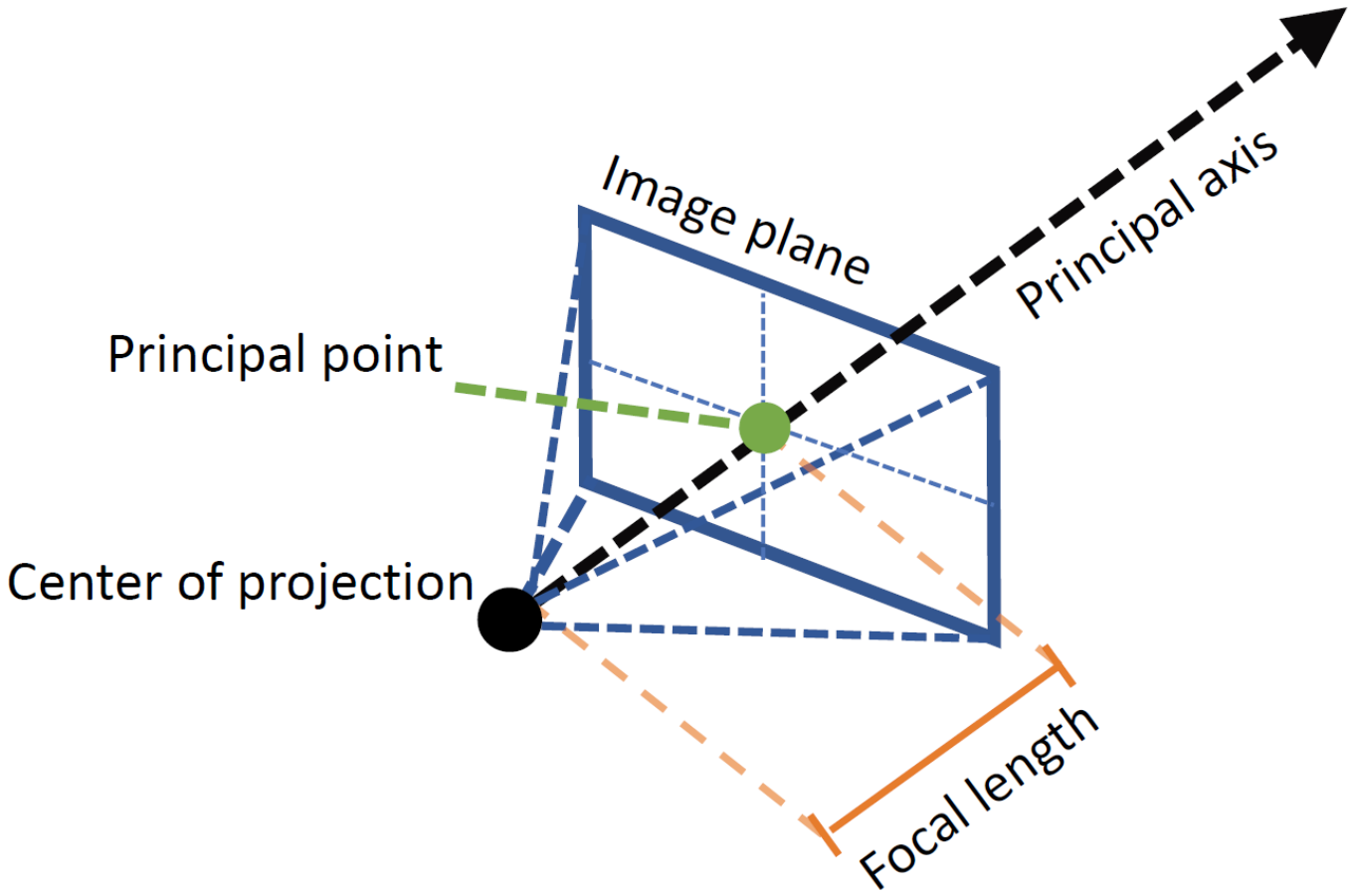
Pinhole camera



Add a barrier to block off most of the rays

- This reduces blurring
- The opening known as the **aperture**
- How does this transform the image?

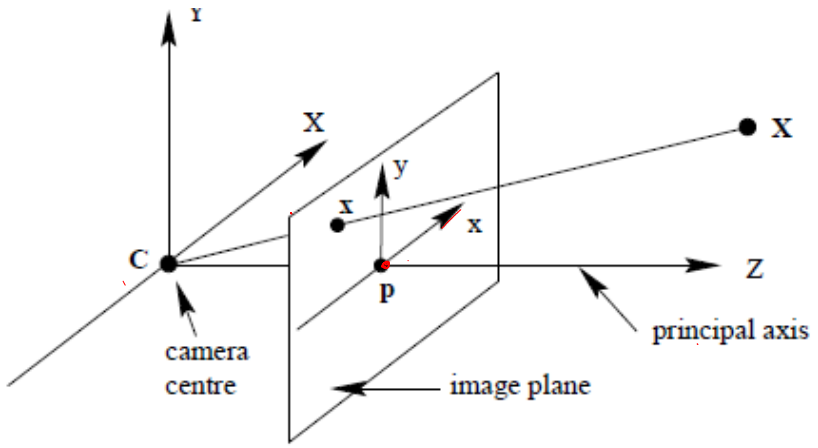
Geometric Model: A Pinhole Camera

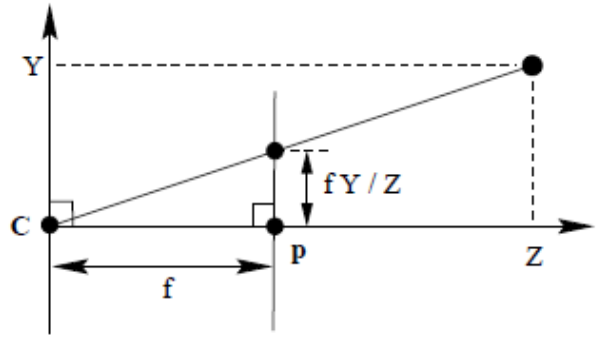


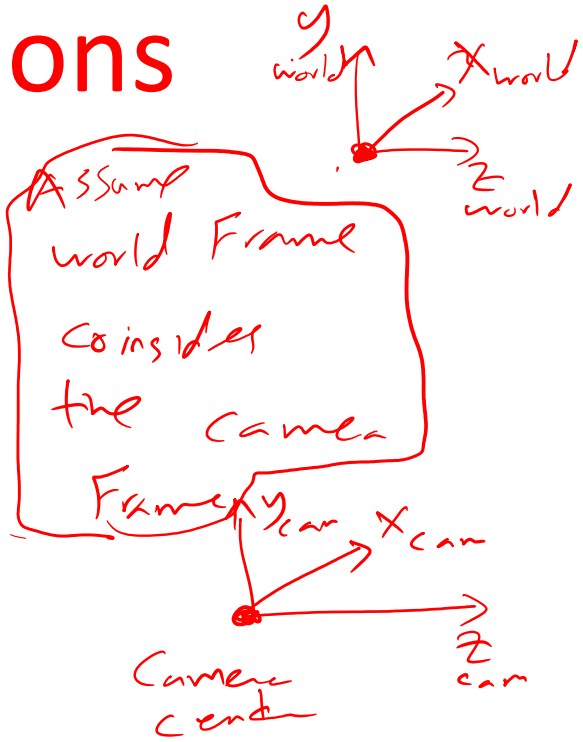
Camera Models

- Starting with a model for the basic pinhole camera, we will develop models for the following cameras:
  1. CCD Imager Pinhole Camera,
  2. Finite Projective Camera,
  3. General Projective Camera,
  4. Orthographic Projective Camera,
  5. Scaled Orthographic Projective Camera, and
  6. Weak Perspective Projective Camera
- All of the above camera models are obtained by placing different kinds of constraints on the structure and the rank of a 3×4 matrix known as the Camera Projection Matrix

The Basic Pinhole Model - Notations





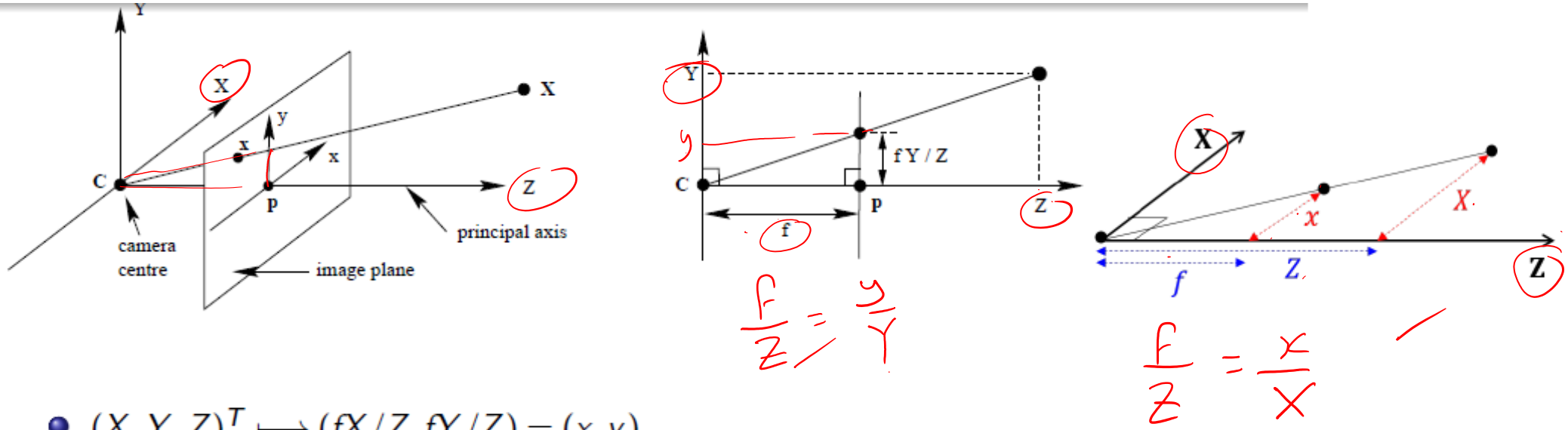


- p : Principle Point
- x: Image point (true image pixel)
- C: Center of Projection (COP) or optical center of the camera
- X: Object point $(X, Y, Z)^T$
- xy−coordinate system in the image plane: Center of the coordinate system of the image plane
  - Measurements usually done with respect to one of the corners not to the center of the sensor plane as shown in the figure above
- f : Focal length of the camera

The Basic Pinhole Model

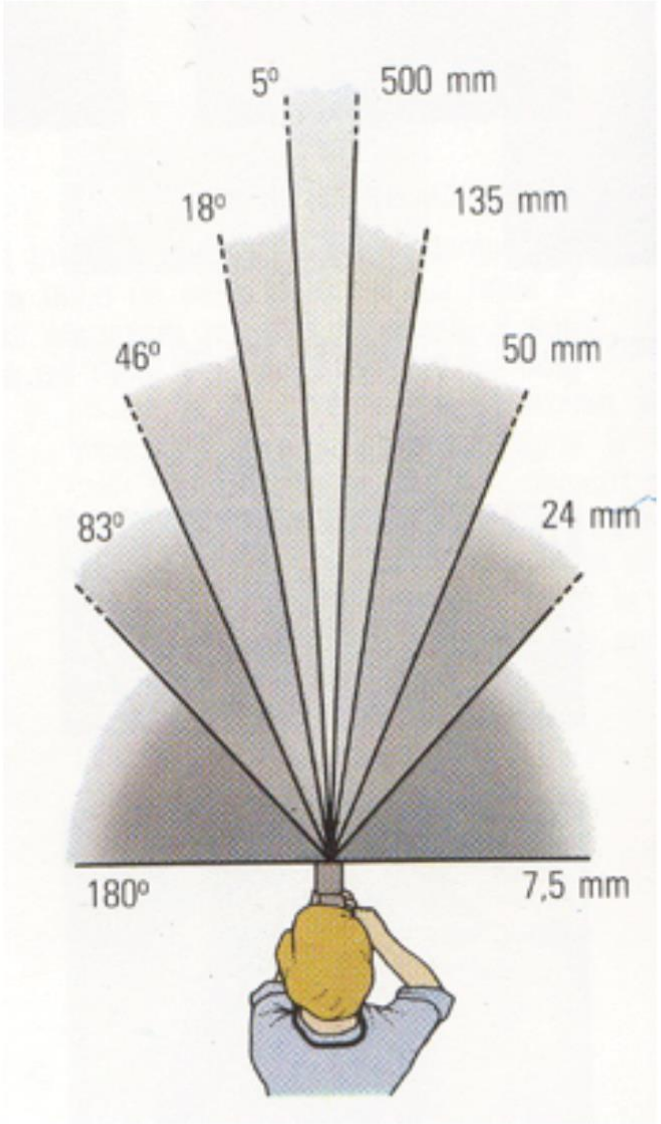
- Comparing similar triangles, the relationship between the world point $(X, Y, Z)^T$ and the corresponding image point (x, y) is given by:
  - $x = fX/Z$, $y = fY/Z$, $z = Z$ (Nonlinear relationships)
- The above relationship is based on measuring the pixel coordinates (x, y) with respect to **the principle point**.
  - More commonly, though, the pixel coordinates are measured with respect to a corner of the image frame.

The Basic Pinhole Model



- $(X,Y,Z)^T \longmapsto (fX/Z, fY/Z) = (x,y)$
  Object Point → Image Point
  - This follows from Triangle Similarity
- $x = fX/Z$ or $(xZ = fX)$, $y = fY/Z$ or $(yZ = fY)$, $z = Z$ (Nonlinear relationships)
- Using homogeneous representation of the image and object (or world) points, we can write the following linear relationship between the pixel coordinates on the left and the corresponding world point coordinates on the right:
  - $$\begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} xz \\ yz \\ z \end{pmatrix} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$
    HC Image Point; HC Object Point

Focal length in practice



24mm



50mm



135mm



The Basic Pinhole Model

- $$\underbrace{\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}}_{\text{HC Image Point}} = \begin{pmatrix} xz \\ yz \\ z \end{pmatrix} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{HC Object Point}}$$
- $$\underbrace{\begin{pmatrix} xz \\ yz \\ z \end{pmatrix}}_{\text{HC Image Point } \vec{x}} = \begin{pmatrix} fX \\ fY \\ Z \end{pmatrix} = \underbrace{\begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{K} \left[\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I} \underbrace{\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}}_{\vec{0}} \right] \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{HC Object Point } \vec{X}}$$
- $\vec{x} = K \left[I | \vec{0} \right] \vec{X}$

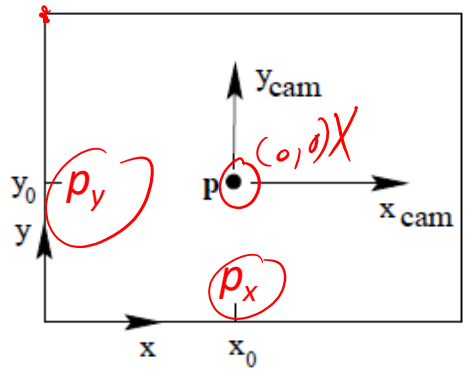
The Basic Pinhole Model

- $\vec{x} = K\left[I|\vec{0}\right]\vec{X}$
- $K = \begin{bmatrix} f & 0 & 0 \\ 0 & f & 0 \\ 0 & 0 & 1 \end{bmatrix}$: Intrinsic parameters of the camera
  - $\left[I|\vec{0}\right]$: Augmented Matrix
- $\vec{x}_{3\times1} = P_{3\times4}\vec{X}_{4\times1}$
  - $P_{3\times4} = K\left[I|\vec{0}\right]$ is the camera matrix

The Basic Pinhole Model

- The world frame shown in the previous slides is in reality The Camera Coordinate Frame - because it is anchored at the camera.
- Therefore, what we have derived should be shown as: $\vec{x} = K \left[I_{3\times 3} | \vec{0} \right] \vec{X_{cam}}$, where $\vec{X}_{cam}$ are the Camera Frame Coordinates of world $\vec{X}$

The Basic Pinhole Model



- As we noted before, in practice, the image coordinates are always measured with respect to one of the sensor corner.
- Thus, let's go back to the previous basic pinhole model and change the coordinate frame of the image plane from the center to the left lower corner.
- Therefore, the principle point (PP) of the camera will be given with respect to the new origin of the coordinate frame of the image plane by (p_x, p_y)
- $(X, Y, Z)^T \longmapsto (fX/Z + p_x, fY/Z + p_y) = (x, y)$
  Object Point — Image Point

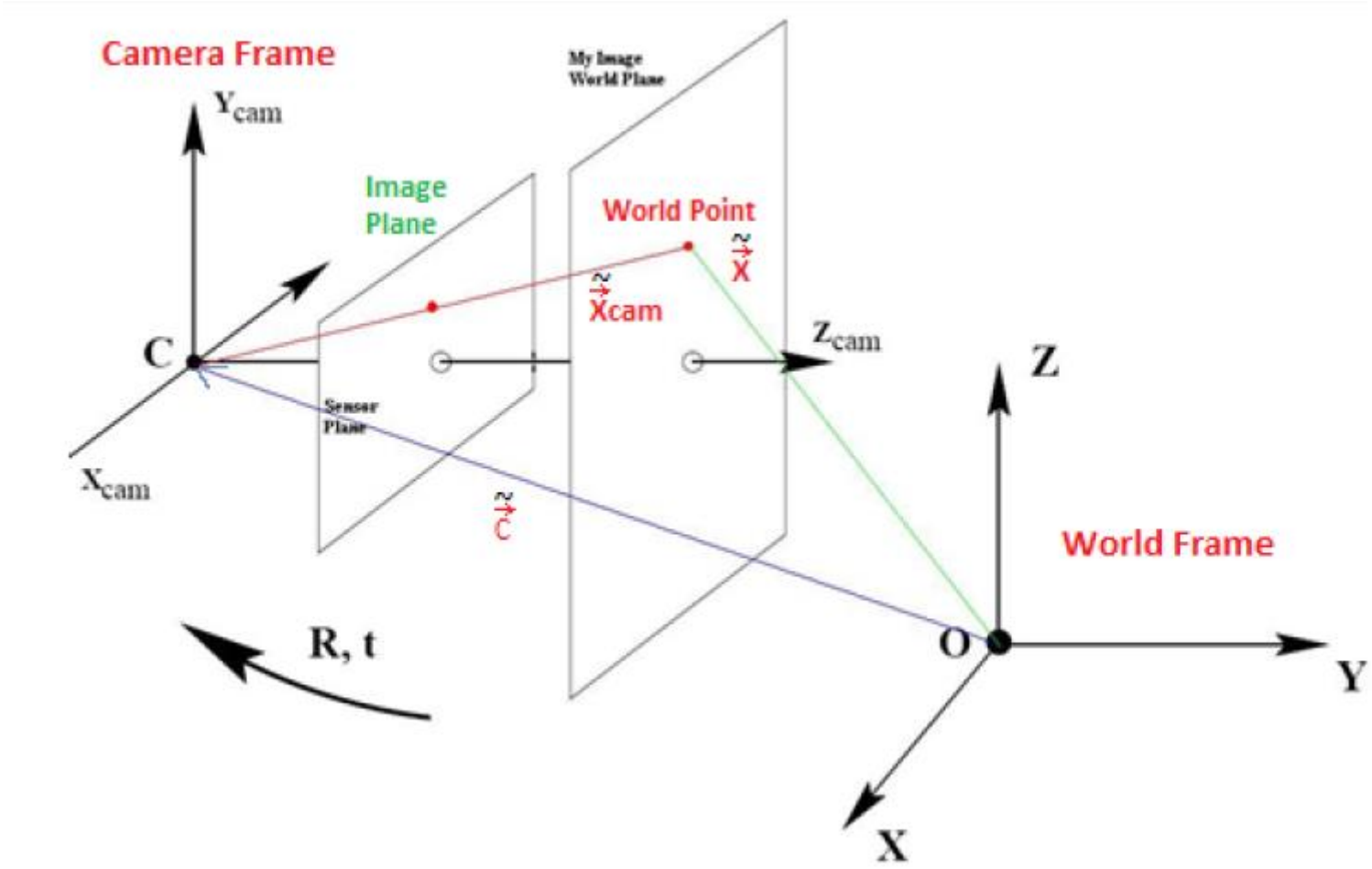
The Basic Pinhole Model

- $$\underbrace{\begin{pmatrix} x \\ y \\ 1 \end{pmatrix}}_{\text{HC Image Point}} = \begin{pmatrix} xz \\ yz \\ z \end{pmatrix} = \begin{pmatrix} fX + p_x Z \\ fY + p_y Z \\ Z \end{pmatrix} = \begin{bmatrix} f & 0 & p_x & 0 \\ 0 & f & p_y & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \underbrace{\begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}}_{\text{HC Object Poi}}$$
  - $x = (fX + p_x Z)/Z$
  - $y = (fY + p_y Z)/Z$
- $\vec{x} = K\left[I|\vec{0}\right]\vec{X}$ or $\vec{x} = K\left[I|\vec{0}\right]\vec{X_{cam}}$
- $K = \begin{bmatrix} f & 0 & p_x \\ 0 & f & p_y \\ 0 & 0 & 1 \end{bmatrix}$: Intrinsic parameters of the camera (or the Camera Calibration Matrix $P_{3\times 4}$)

Relationships between the camera coordinate frame and the world frame

- Let's now separate the camera coordinate frame and the world frame
- In order to find the relationship between $\vec{X}_{cam}$ and $\vec{X}$, in camera modeling work, it is often best to first establish the relationship between physical coordinate that we denote by $\tilde{\vec{X}}_{cam}$ and $\tilde{\vec{X}}$

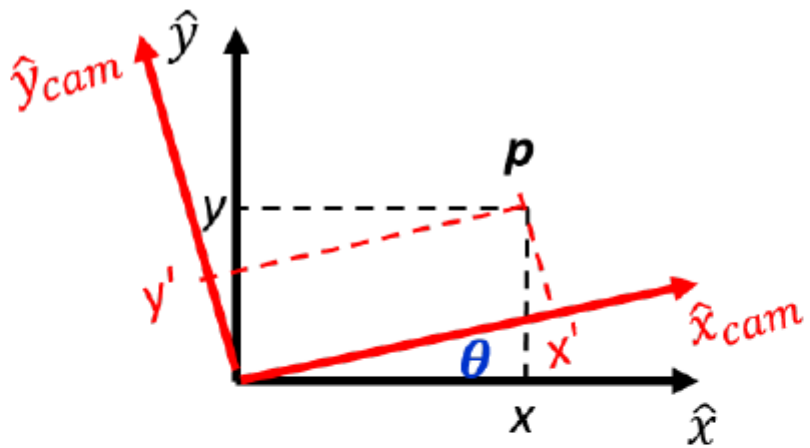
Relationships between the camera coordinate frame and the world frame



Relationships between the camera coordinate frame and the world frame

- Let R denote the rotation from the physical world coordinates to the camera frame coordinates.
  - To understand that what we mean by 'from' and 'to' clauses, assume that the two origins are at exactly the same point.
  - Our definition of R would imply that $\tilde{\vec{X}}_{cam} = R\tilde{\vec{X}}$.
  - To see this more clearly, consider the 2D case at the next slide:

Relationships between the camera coordinate frame and the world frame



- $\tilde{\vec{X}}_{cam} = \begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{bmatrix} cos\theta & sin\theta \\ -sin\theta & cos\theta \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = R\tilde{\vec{X}}.$
- So, when R multiplies the coordinates of a point in the world frame, we get the coordinates of the **same point** in the camera frame.

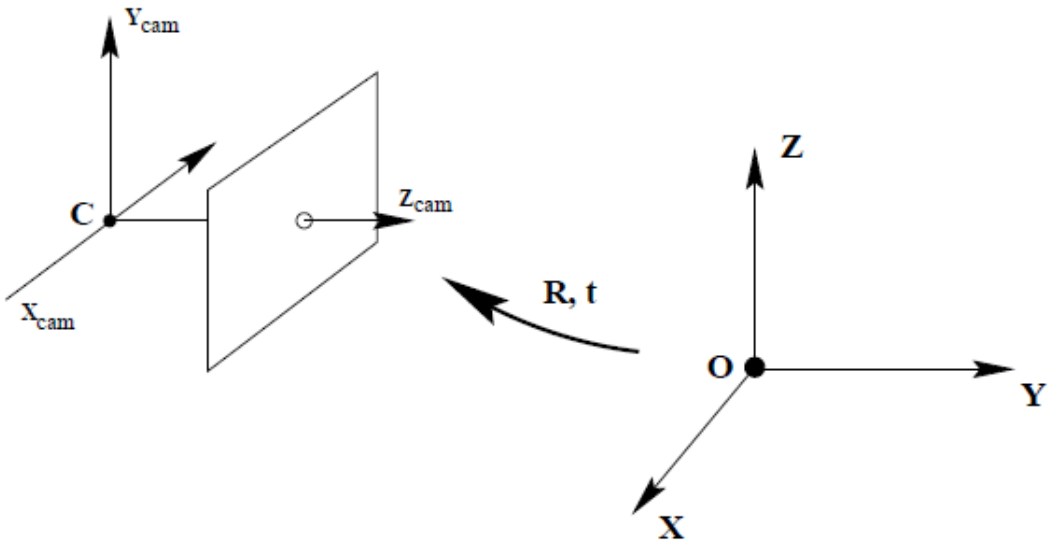
Extrinsic Parameters of the Camera

- Let's now denote the relationships between the camera coordinate frame and the world frame by the extrinsic parameters of the camera.

Extrinsic Parameters of the Camera

- Let $\tilde{C}$ denote the transnational vector from the world frame origin to the camera frame origin.
- If we ignore the rotational misalignment between the two frames, we have $\tilde{\vec{X}}_{cam} = \tilde{\vec{X}} - \tilde{\vec{C}}$.
  - When we have both a translational $\tilde{\vec{C}}$ and a rotation R from the world frame to the camera frame, we have $\tilde{\vec{X}}_{cam} = R\left(\tilde{\vec{X}} - \tilde{\vec{C}}\right)$.
    - For the world point $\tilde{\vec{X}}$, the subtraction $\left(\tilde{\vec{X}} - \tilde{\vec{C}}\right)$ gives us the coordinates of the same world point in the camera frame at a vector distance $\tilde{\vec{C}}$ but with the camera frame still aligned with the world frame. That by multiplying this by R should yield $\tilde{\vec{X}}_{cam}$ follows from our previous explanation of how R works.

Extrinsic Parameters of the Camera



- Camera rotation and translation with respect to the world frame that is shown to the right of the figure above.
- $\widetilde{X}_{cam} = R(\widetilde{X} - \widetilde{C})$
  - $\widetilde{X}_{cam}$ and $\widetilde{X}$ are in physical coordinates
  - Do translation first and then do the rotation part

Extrinsic Parameters of the Camera

- The equation $\tilde{\vec{X}}_{cam} = R\left(\tilde{\vec{X}} - \tilde{\vec{C}}\right)$ can be adapted to the homogeneous coordinates by:
- $$\vec{X}_{cam} = \begin{pmatrix} \tilde{\vec{X}}_{cam} \\ 1 \end{pmatrix} = \begin{bmatrix} R & -R\tilde{\vec{C}} \\ 0 & 1 \end{bmatrix} \begin{pmatrix} \tilde{\vec{X}} \\ 1 \end{pmatrix} = \begin{bmatrix} R & -R\tilde{\vec{C}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} =$$
  $$\begin{bmatrix} R & -R\tilde{\vec{C}} \\ 0 & 1 \end{bmatrix} \vec{X}$$
- Substituting the relationship between the homogeneous vectors $\vec{X}_{cam}$ and $\vec{X}$ in the pinhole camera model shown before, we can write for the pixel coordinates:
  - $\vec{x} = K\,[I|\vec{0}]\,\vec{X}_{cam}$
  - $\vec{x} = K\,[I|\vec{0}] \begin{bmatrix} R & -R\tilde{\vec{C}} \\ \vec{0}^T & 1 \end{bmatrix} \vec{X}$

Extrinsic Parameters of the Camera

- $\vec{x} = K\,[I|\vec{0}] \begin{bmatrix} R & -R\tilde{\vec{C}} \\ \vec{0}^T & 1 \end{bmatrix} \vec{X}$
- Note: $\left[I_{3\times 3}|\vec{0}\right] \begin{bmatrix} R & -R\tilde{\vec{C}} \\ \vec{0}^T & 1 \end{bmatrix} =$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & (-R\tilde{\vec{C}})_x \\ r_{21} & r_{22} & r_{23} & (-R\tilde{\vec{C}})_y \\ r_{31} & r_{32} & r_{33} & (-R\tilde{\vec{C}})_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} & (-R\tilde{\vec{C}})_x \\ r_{21} & r_{22} & r_{23} & (-R\tilde{\vec{C}})_y \\ r_{31} & r_{32} & r_{33} & (-R\tilde{\vec{C}})_z \end{bmatrix}$$

- $= \left[R|-R\tilde{\vec{C}}\right] = R\left[I|-\tilde{\vec{C}}\right]$

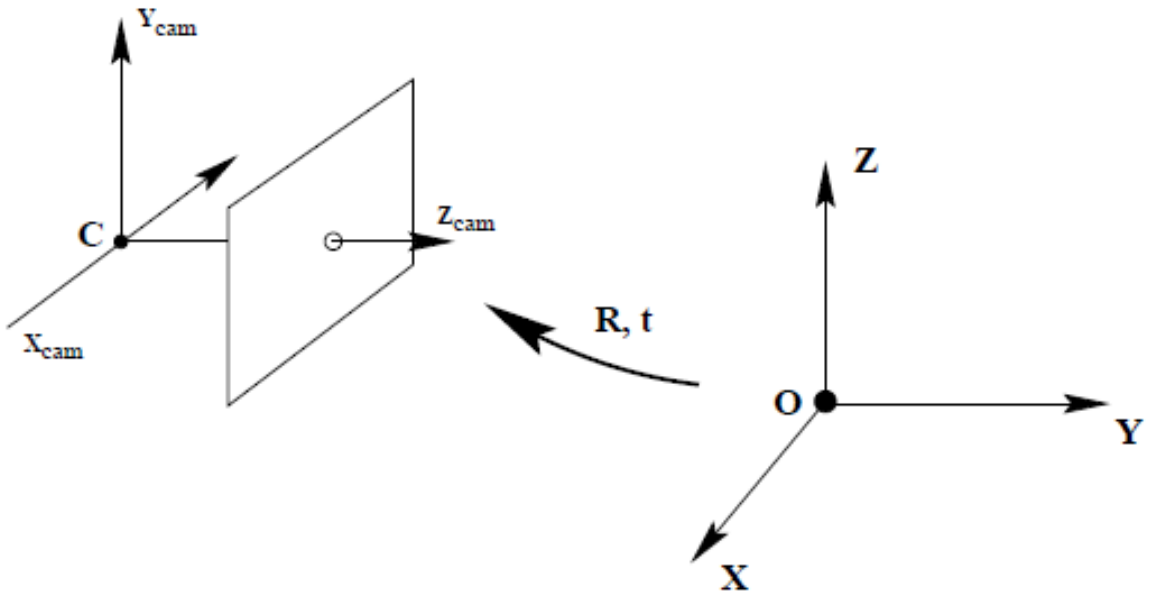
Extrinsic Parameters of the Camera

- Thus, $\vec{x} = K\left[R|-R\tilde{\vec{C}}\right]\vec{X} = KR\left[I|-\tilde{\vec{C}}\right]\vec{X} = \boldsymbol{K[R|t]}\vec{X} = \boldsymbol{P}\vec{X}$

  - $\vec{x}$: pixel coords (homogeneous)
  - K: camera parameters (intrinsic)
  - R: World-to-Camera rotation (physical)
  - $\tilde{\vec{C}}$: World-to-Camera Translation (physical)
  - $\vec{t} = -R\tilde{\vec{C}}$
  - $\vec{X}$: World point (homogeneous)

The equation above will serve as our fundamental equation for modeling a pinhole camera.

Extrinsic Parameters of the Camera



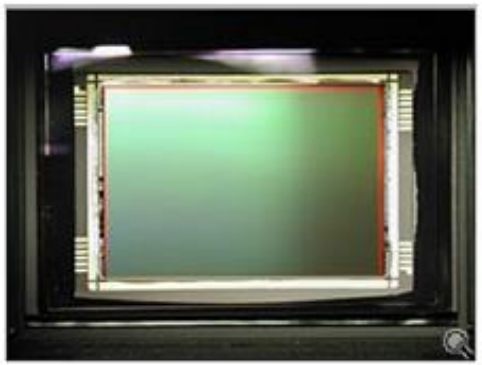
- $\vec{x} = K\,[I|\vec{0}] \begin{bmatrix} R & -R\tilde{C} \\ 0 & 1 \end{bmatrix} \vec{X} = K\left[R|-R\tilde{C}\right]\vec{X} = KR\left[I|-\tilde{C}\right]\vec{X} = P\vec{X}$
  - $P = KR\left[I|-\tilde{C}\right] = K\left[R|-R\tilde{C}\right] = K\,[R|t]$
    - $t = -R\tilde{C}$

Taking into account the sampling of the image plane

- For images recorded with digital cameras (e.g. CCD cameras), we must also factor in the sampling rates in the image plane.
- Let's start by assuming the that the 2D array of photo sensors in the image plane is perfectly rectangular, with m_x cells per unit length along x and m_y cells per unit length along y.

Taking into account the sampling of the image plane





- So recall that we assumed the CCD image sensor have m_x elements per unit length along x and m_y elements per unit length along y.
- We may now associate the "**discretized**" location (I, J) with the pixel at (x, y), with:
  - $I = \boldsymbol{m_x} x$ and $J = \boldsymbol{m_y} y$
    - Note 1: I and J are known as the pixel coordinates in image **(True pixels)**
    - Note 2: Obviously, the products $m_x x$ and $m_y y$ do not by themselves yield integer values . However, to get integers indices associated with the cells, all we have to do is to ignore the fractional parts of $m_x x$ and $m_y y$.

Taking into account the sampling of the image plane

- Using our foundational equations for x and y that we have discussed before in the the Basic Pinhole Model, we can write:
  - $I = m_x x = m_x(fX/Z + p_x) = m_x fX/Z + \boldsymbol{m_x p_x}$
  - $J = m_y y = m_x(fY/Z + p_y) = m_y fY/Z + \boldsymbol{m_y p_y}$
- If we use $\boldsymbol{x_0 = m_x p_x}$ and $\boldsymbol{y_0 = m_y p_y}$ as the integer coordinates of the principle point in the image plane, we can write:
  - $I = m_x fX/Z + \boldsymbol{x_0}$ and $J = m_y fY/Z + \boldsymbol{y_0}$
  - $I = \frac{m_x fX + x_0 Z}{Z}$ or $\boldsymbol{IZ = m_x fX + x_0 Z}$, and $J = \frac{m_y fY + y_0 Z}{Z}$ or $\boldsymbol{JZ = m_y fY + y_0 Z}$
  - These last two equations can be turned into the following composite relationship between the homogeneous representations of the "discrete" pixel coordinates and the world point in the camera coordinate frame as shown in the next slide.

Taking into account the sampling of the image plane

- $$\begin{pmatrix} I \\ J \\ 1 \end{pmatrix} = \begin{pmatrix} IZ \\ JZ \\ Z \end{pmatrix} = \begin{pmatrix} m_x f X + x_0 Z \\ m_y f Y + y_0 Z \\ Z \end{pmatrix} =$$

$$\begin{bmatrix} m_x f & 0 & x_0 & 0 \\ 0 & m_y f & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & x_0 & 0 \\ 0 & \alpha_y & y_0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} X \\ Y \\ Z \\ 1 \end{bmatrix}$$

  - with $\alpha_x = m_x f$, $\alpha_y = m_y f$, and (x_0, y_0): the principle point is in image plane
  - You can think of:
    - α_x as the focal length in terms of the number of pixels using the x-sampling rate (or x digitization), and of
    - α_y the focal length in terms of the number of pixels using the y-sampling rate (of y digitization).

Taking into account the sampling of the image plane

- $\begin{pmatrix} IZ \\ JZ \\ Z \end{pmatrix}$ is obviously the homogeneous representation of the discrete pixel coordinates (I, J).
  - Going forward, we will represent this homogeneous vector by $\vec{x}$.
- So we again have:
  - $\vec{x} = K \left[I_{3\times 3} | \vec{0} \right] \vec{X}_{cam}$, with $K = \begin{bmatrix} m_x f & 0 & x_0 \\ 0 & m_y f & y_0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$
  - This relationship between the homogeneous vectors $\vec{x}$ and $\vec{X}_{cam}$ is the same as what we saw before. **The only difference here is the form of the intrinsic parameters of the camera calibration matrix K.**

Taking into account the sampling of the image plane

- Therefore, our overall relationship between a world point $\vec{X}$ and the corresponding pixel $\vec{x}$ remains the same as shown earlier except for the intrinsic calibration matrix K being as shown in the previous slide.

- Thus, we produce below one of those two relationship:
  - $\vec{x} = \underbrace{KR\left[I_{3\times3}|-\tilde{\vec{C}}\right]}_{\text{Projection Matrix P}}\vec{X} = P_{3\times4}\vec{X}$
  - Note: $P_{3\times4}$ is homogeneous and $K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}$

Camera Matrix: Degree of Freedoms

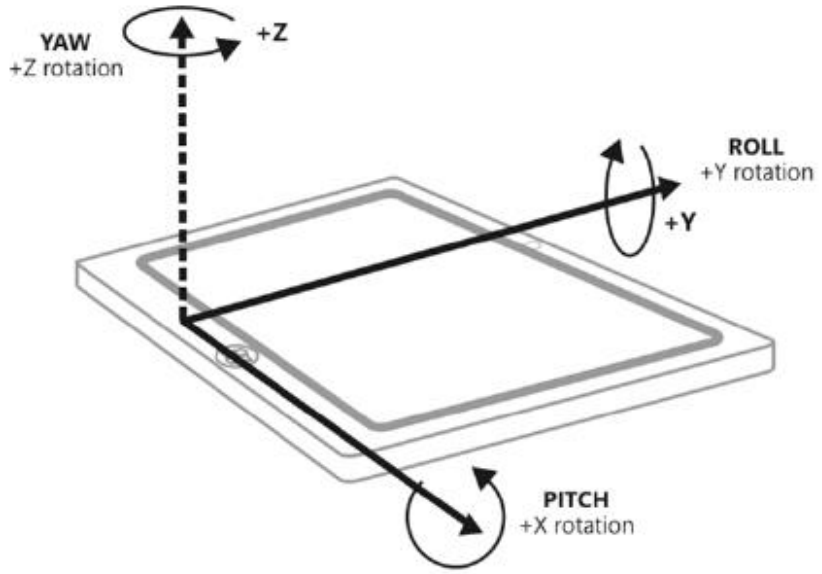
- $P_{3\times4} = KR\left[I_{3\times3}|-\tilde{\vec{C}}\right] = K\left[R|-R\tilde{\vec{C}}\right] = K_{3\times3}\left[R_{3\times3}|\vec{t}_{3\times1}\right]$
- **The 3×4 projection matrix P has 10 Degrees of Freedom (DoF).**
  - Of these 10 DoF: K accounts for 4, R for 3, and $\tilde{\vec{C}}$ for the remaining 3.
  - $K = \begin{bmatrix} \alpha_x & 0 & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times3}$, 4 DoF
  - $R_{3\times3}$, Euler angles: yaw, pitch, and roll. 3 DoF
  - $t_{3\times1} = -R\tilde{\vec{C}}$, 3 DoF
  - Total: **4 DoF + 3 DoF + 3 DoF = 10 DoF**
- But $P_{3\times4} = 12$ parameters, thus **we should have 11 DoF.** Therefore **we are missing 1 DoF**.

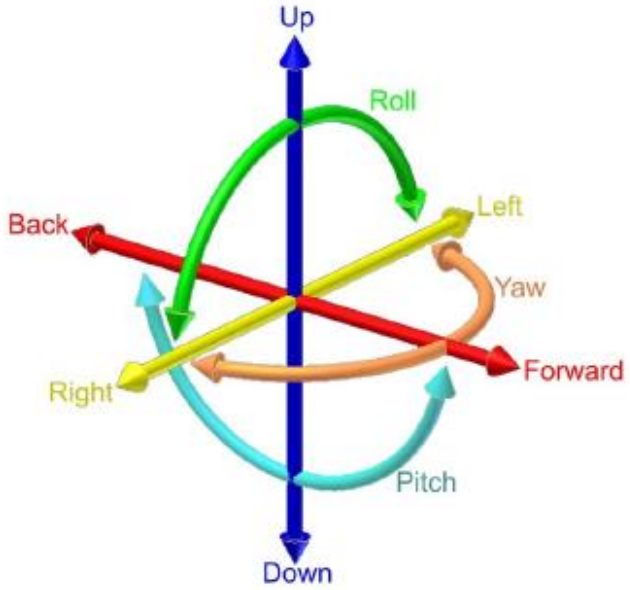
Aside: Euler Angles

- $R_{3\times3} = R_x(\gamma)R_y(\beta)R_z(\alpha) =$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\gamma & -sin\gamma \\ 0 & sin\gamma & cos\gamma \end{bmatrix} \begin{bmatrix} cos\beta & 0 & sin\beta \\ 0 & 1 & 0 \\ -sin\beta & 0 & cos\beta \end{bmatrix} \begin{bmatrix} cos\alpha & -sin\alpha & 0 \\ sin\alpha & cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

  - Euler angles: yaw, pitch, and roll.





Camera Matrix: Degree of Freedoms

- Now consider that since the 3×4 P has a maximum of 11 DoF, as to what might be the practical significance of the remaining single DoF in P.
- Consider the following form for K:

Camera Matrix: Degree of Freedoms

- Missing DoF
- $K = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$
  - 5 DoF
  - s : skew factor as a result of skew coordinate (rectangular representation). Will be called the skew parameter.
- With this K, the projection matrix P will have the maximum allowable DoF of 11.
  - **As it turns out, the additional parameter s in K is required if the image sensor array is not perfectly rectangular.**
  - In digital cameras, there is always a bit of "skew" in the xy-layout of the sensor cells.

General Notes

- K matrix is non-singular and will never be singular
- R matrix is by definition non-singular or it won't be a rotation matrix
- $\vec{t}$ vector can go to zero (no translation)

Recall: The Pinhole Camera Model

- $P = KR\left[I|-\tilde{C}\right] = K\left[R|-R\tilde{C}\right] = K[R|t]$
  - $K = \begin{bmatrix} \alpha_x & s & x_0 \\ 0 & \alpha_y & y_0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$
  - $s = 0$: Ideal Camera

Finite Projective Cameras

- In general, and specially if one allows for virtual cameras, the relationship between a world point $\vec{X}$ and its corresponding pixel at $\vec{x}$ is expressed as:
- $\vec{x}_{3\times1} = \boldsymbol{P}_{3\times4}\vec{X}_{4\times1}$
  - In general , we will express $\boldsymbol{P}$ as $P = \begin{bmatrix} \vec{p}_1 & \vec{p}_2 & \vec{p}_3 & \vec{p}_4 \end{bmatrix}$ when we need to identify its four columns separately.
  - And, sometimes, we will show $\boldsymbol{P}$ as $P = [M|\vec{p}_4]$, where M is the 3×3 submatrix formed by the first three columns of P.
- **We now claim that any 3×4 real matrix whose first three columns constitute a non-singular submatrix is a valid camera.**
  - We refer to the set of all such cameras as **Finite Projective Cameras**.

Notes on the Finite Projective Cameras

- $P = KR\left[I|-\widetilde{C}\right] = M\left[I_{3\times 3}|M^{-1}\vec{p}_4\right] = [M|\vec{p}_4]$
  - $M = KR$, and $\widetilde{C} = -M^{-1}\vec{p}_4$
- For pinhole and finite projective cameras the left 3×3 sub-matrix of P is non-singular.

Finite Projective Cameras: Summary

- Obviously, all digital cameras are examples of Finite Projective Cameras. However, the **Finite Projective Set** includes cameras that cannot be realized physically but that may nonetheless be useful as **virtual cameras**.
- When a Finite Projective Camera cannot be realized physically, it may be:
  - on account of where the COP is in relation to the object, or
  - on account of where the image is formed on the image plane, or
  - on account of the values given to the intrinsic parameters of the camera.

General Projective Camera

- The Finite Projective Camera is obviously a more general category than the pinhole camera (with and without the skew parameter).
- An even more general category of cameras is **the General Projective Camera**.
  - **The only constraint we now place on the projection matrix P is that its rank be exactly 3.**

$$P = K[R \mid \vec{t}] = [\vec{p_1}\ \vec{p_2}\ \vec{p_3}\ \vec{p_4}] = [M_{3\times3} \mid \vec{p_4}], \quad M = KR, \quad \vec{p_4} = K\vec{t}$$

Augment matrix. ← 4 columns

finite projective camera ← $P_{3\times4}$ = P has a Rank = 3 & M is non-singular → Rank(M) = 3

$$\tilde{C} = -R^{-1}\vec{t}$$

General projective camera ← $P_{3\times1}$ just has a Rank = 3, $M_{3\times3}$ can be Singular

if this is the case our camera is not physical, it is called Virtual Camera.

M has a Null space ← Let that null space be $\vec{d}_{3\times1}$

Camera Location $\equiv \tilde{C} = \vec{d}$, at infinity



General projective camera

Affine camera

Orthographic camera

Scaled orthographic camera

Weak perspective camera

$$P_{3\times4} = \underbrace{\left[\,M \mid \vec{P_4}\,\right]}_{\text{singular}} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & P_{14} \\ m_{21} & m_{22} & m_{23} & P_{24} \\ m_{31} & m_{32} & m_3 & P_{34} \end{bmatrix} \rightarrow \text{make this } [0\ 0\ 0\ 1]$$

$$P_{3\times4} = \begin{bmatrix} m_{11} & m_{12} & m_{13} & P_{14} \\ m_{21} & m_{22} & m_{23} & P_{24} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P_{3\times4} = \left[\begin{array}{ccc|c} 1 & 0 & 0 & P_{14} \\ 0 & 1 & 0 & P_{24} \\ 0 & 0 & 0 & 1 \end{array}\right], \quad m_{11}=1,\ m_{22}=1 \text{ \& the rest of } m\text{'s} = 0$$

$$\vec{i} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \quad \vec{c} = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

$$P_{3\times4} = \lambda \left[\begin{array}{ccc|c} 1 & 0 & 0 & P_{14} \\ 0 & 1 & 0 & P_{24} \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \lambda = \text{isotropic scaling factor}$$

$$= \left[\begin{array}{ccc|c} \lambda_1 & 0 & 0 & P_{14} \\ 0 & \lambda_2 & 0 & P_{24} \\ 0 & 0 & 0 & 1 \end{array}\right], \quad \lambda_1 \text{ \& } \lambda_2 \text{ are different scaling factors.}$$

Camera at Infinity

- Despite the sense conveyed by "infinity" in the title of this slide, these cameras are of great practical importance when images are formed under orthographic and weak perspective conditions.
- Recall:
  - **Any 3×4 matrix is a projective camera provided its rank is 3.**
  - A general projective camera (that is NOT finite projective) has the following property:
    - The submatrx M of $P = [M|\vec{p}_4]$ is singular and that the null vector $\vec{d}$ of M gives us the camera center $\vec{C} = \begin{pmatrix} \vec{d} \\ 0 \end{pmatrix}$ at infinity for such a camera.

Camera at Infinity

- A general projective camera located at infinity.
  - Has parallel beams of rays parallel to either x-direction (world X), or y-direction (world Y) or z-direction (world Z).
- We will now consider a sub-category of such general projective cameras - the sub-category (or Special Case) is known as the Affine Camera

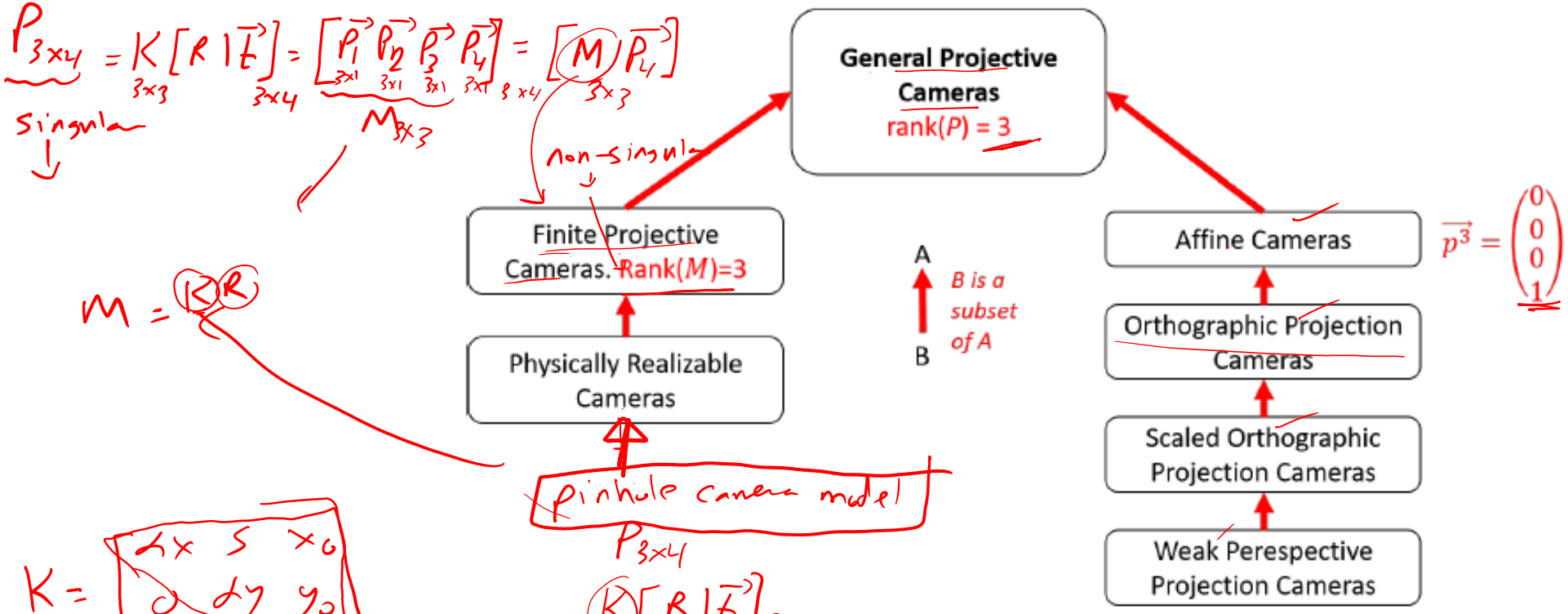
Affine Cameras

- Under Affine Cameras, we will consider:
  - Orthographic Projection Camera (rays are parallel),
  - Scaled Orthographic Projection Camera, and
  - Weak Perspective Projection Camera
- In general, **a camera is affine if the last row of the 3×4 P is** $\begin{pmatrix} 0 & 0 & 0 & 1 \end{pmatrix}$.
  - This means the principle plane π_∞ which is the same as plane at infinity is given by $\begin{bmatrix} 0 & 0 & 0 & 1 \end{bmatrix}^T$
  - Affine camera maps parallel lines to parallel lines
  - Obviously, with $P = [M|\vec{p}_4]$, M for an affine camera is given by

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ 0 & 0 & 0 \end{bmatrix}$$

Important Note: Moving Forward

- You will see that all Central Projection Cameras of practical interest can be mapped to the following hierarchy of cameras:



Note: *All of these different types of cameras can be used for Computational Photography*

Orthographic Projection Camera

- In its generic form, this affine camera (i.e. the Orthographic camera which is a special case of an affine camera.) is defined by:
  - $P = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $M = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, $\vec{p}_4 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$
  - Since $\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$ is the null vector of M in this case, the CoP of the camera is at $\begin{bmatrix} \vec{d} \\ 0 \end{bmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}$, which is a point at ∞ along the world Z-axis, this camera gives us an orthographic projection of the object onto the xy-plane.
  - By orthographic, we mean projection with **a bundle of parallel rays**.
    - Obviously, only those object points would be projected that are visible to the camera.

Orthographic Projection Camera

- For other poses of the camera, we introduce a rotation R and a translation $\vec{t}$ from the world frame to the camera frame.
- Following the same logic as we did before at the beginning of these slides, we can write:
  - $\vec{x} = P\vec{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} R & \vec{t} \\ \vec{0}^T & 1 \end{bmatrix} \vec{X} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{r}_3^T & t_3 \\ \vec{0}^T & 1 \end{bmatrix} \vec{X}$, where $\vec{r}_1^T$, $\vec{r}_2^T$, and $\vec{r}_3^T$ are the three row vectors of R, and where t_1, t_2, t_3 are the three components of $\vec{t}$.
  - Therefore, the projection matrix P of an orthoprojection camera at pose $(R, \vec{t})$ with respect to the world frame is given by $P = \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1 \end{bmatrix}$.
    - By World frame here, we mean the coordinate frame in which the generic configuration is defined.

Scaled Orthographic Projection Camera

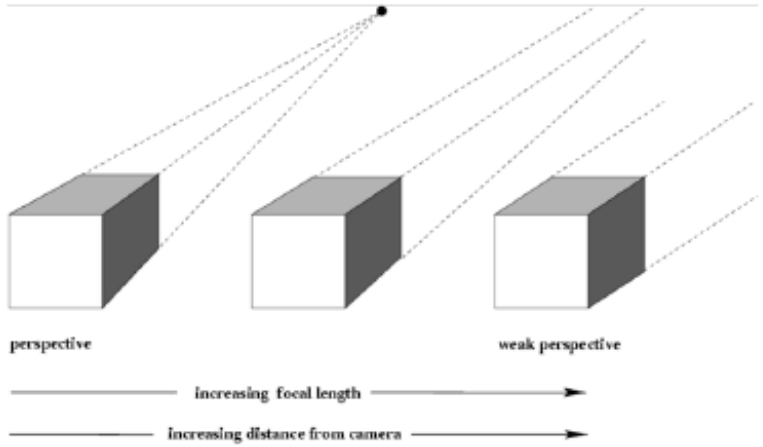
- A Scaled Orthographic Projection Camera is a regular Orthographic Projection Camera as described before that is followed by isotropic scaling of the image.
  - This means that whatever image is produced by the P matrix of the previous slide needs to be scaled by a factor k.
  - Therefore, the new P would be given by:
    - $$P = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1 \end{bmatrix} = \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1/k \end{bmatrix}.$$
    - Note: for both the P here and in the previous case, the first two rows of M submatrix of P must be orthogonal.

Weak Perspective Projection Camera

- This camera differs from the one in the previous case in only one sense:
  - We now use two different scale factors for the x and the y directions in the image plane.
  - Consequently, we can write the projection matrix as:
  - $$P = \begin{bmatrix} \alpha_x & 0 & 0 \\ 0 & \alpha_y & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \vec{r}_1^T & t_1 \\ \vec{r}_2^T & t_2 \\ \vec{0}^T & 1 \end{bmatrix}.$$
    - Note: The first two rows of M submatrix of P would still be orthogonal.

Important Notes

- The last two affine camera models (Scaled and Weak-Perspective Cameras) are often used for images of distant objects recorded with long focal length lenses and when the objects subtend small angles at the camera center.







Imaging of Various Geometrical Entities:
Action of Projective Camera on Point

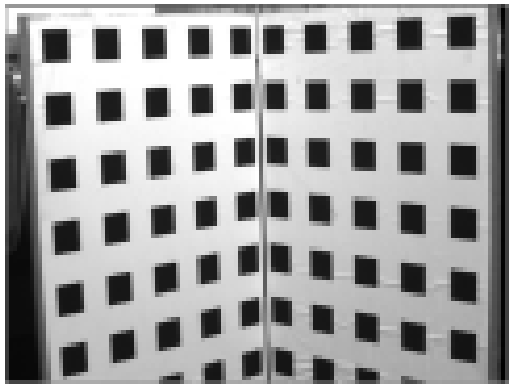
- Forward projection
- $\vec{x}_{1\times 3} = P_{3\times 4}\vec{X}_{1\times 4} = [M|\vec{p}_4]\vec{X}_{1\times 4} = \begin{bmatrix} \vec{p}_1 & \vec{p}_2 & \vec{p}_3 & \vec{p}_4 \end{bmatrix}\vec{X}_{1\times 4}$
  - Back-projection
    - $\vec{X} = P^{\dagger}\vec{x}$ — Line
    - $P^{\dagger}$:pseudo-inverse $P^{\dagger} = P^T(PP^T)^{-1}$ — not unique.
    - $PP^{\dagger} = I$

Imaging of Various Geometrical Entities

Geometrical Entity	Forward Projection	Backward Projection
Points	3D world points $\rightarrow$ 2D image points	2D image point $\rightarrow$ 3D line
Lines	3D world line $\rightarrow$ 2D image line	2D image line $\rightarrow$3D plane
Planes	3D world plane $\rightarrow$ homography	Does not make sense to apply
Conics	Does not make sense to apply	2D conic $\rightarrow$ cone
Quadrics	Quadric $\rightarrow$ conic	Does not make sense to apply
Absolute Conic	Image of absolute conic Ω_∞ is $K^{-T}K^{-1}$	Does not make sense to apply

- The fact that the Image of absolute conic Ω_∞ is $K^{-T}K^{-1}$ is used in camera calibration.

Camera Calibration (Will cover it if time allows)



- Main goal is to compute the intrinsic parameters K of the camera and the extrinsic parameters $R \& \vec{t}$
  - Computer must know the intrinsic parameters K of the camera and the extrinsic parameters $R \& \vec{t}$ as fast as possible (instantly) in many applications.

Questions

The End